Designing Data-Intensive Applications — Interview Edition

How to use this edition

The night-before version of the DDIA study notes: every chapter's key ideas, comparison tables, likely questions and traps, with the explanations stripped out. Everything here comes from the full chapter notes (`../chapters/NN-*/README.md`); go there when a line needs its "why".

- **Must-know** — the terms and rules you must be able to say in one breath.
- **Key facts & numbers** — comparison tables (LSM vs B-tree, encodings, replication, partitioning, isolation levels, consistency models, consensus, batch vs stream) and the book's numbers.
- **Code to remember** — only the few snippets worth writing on a whiteboard.
- **In the interview** — the most likely questions with 1–3 sentence answers. Say the answer out loud before reading it.
- **Gotchas** — the traps: "wrong idea → right idea".
- **TL;DR** — the chapter in one or two lines.

Suggested passes. 30 minutes: Top 25 questions + every TL;DR + Last-minute checklist. 2 hours: all Key facts tables + In the interview. Before a system-design round, reread Chapters 5, 6, 7, 9 and 11 — they carry most of the questions.

How to answer a DDIA-style question. Name the trade-off, not the product: what does the design guarantee, what does it give up, and what happens when a node, a clock or the network fails?

Top 25 questions across the book

#	Question	One-line answer	Ch
1	Fault vs failure?	A component deviating from spec vs the whole system stopping service; design fault tolerance so faults don't become failures.	1
2	Why percentiles, not averages?	The mean hides the tail; p99/p999 hit the most valuable users and are amplified by fan-out.	1
3	Twitter timeline: fan-out on read or write?	Write fan-out because reads are ~100× writes; hybrid read-time merge for celebrities.	1
4	Relational, document or graph?	Trees loaded whole → document; many-to-many and joins → relational; everything connected → graph.	2
5	LSM-tree or B-tree?	LSM: sequential writes, high write throughput, compaction cost; B-tree: in-place pages, fast predictable reads. Benchmark.	3
6	Why column storage for analytics?	Read only the few columns a query needs; columns compress well and process vectorised.	3
7	Backward vs forward compatibility?	New code reads old data vs old code reads new data; both needed for rolling upgrades.	4
8	Protobuf vs Avro schema evolution?	Protobuf: numeric field tags, never reuse; Avro: writer's vs reader's schema resolved by name, defaults.	4
9	Sync vs async replication?	Sync: follower guaranteed current but can block writes; async: fast but may lose confirmed writes on failover.	5
10	What goes wrong in failover?	Lost writes, reused keys, split brain, bad timeouts.	5
11	Quorums: does w + r > n give strong consistency?	Overlap makes fresh reads likely, but sloppy quorums, LWW, concurrency and partial failures break it — not linearizable.	5, 9
12	Replication-lag anomalies?	Read-your-writes, monotonic reads, consistent prefix — fix by leader reads, replica pinning, causal partitioning.	5
13	Key-range vs hash partitioning?	Range: cheap scans, hot-spot risk; hash: even load, no range queries.	6
14	Why not hash mod N?	Changing N moves most keys; use fixed partitions, dynamic splits, or partitions per node.	6
15	Local vs global secondary index?	Local: cheap writes, scatter/gather reads; global: one-partition reads, async multi-partition writes.	6
16	What does ACID really mean?	Abortability, app invariants, isolation (often not serializable), durability; C is the app's job.	7
17	Snapshot isolation vs serializable?	SI (MVCC) stops dirty reads and read skew but allows write skew and phantoms; only serializable stops all.	7
18	Three ways to serializability?	Serial execution, 2PL, SSI — simple/one core vs blocking/deadlocks vs optimistic/aborts.	7
19	How is a leader wrong about its lease?	Process pauses (GC, VM suspend); fix with fencing tokens checked by storage.	8
20	Why not order events by timestamps?	Clocks skew and jump; LWW silently drops causally later writes; use logical clocks.	8

#	Question	One-line answer	Ch
21	Linearizability vs serializability?	Single-object real-time recency vs transactions in some serial order; both = strict serializability.	9
22	CAP theorem?	During a partition choose linearizable or available; "2 of 3" is misleading; the real cost is latency.	9
23	2PC and its weakness?	Prepare → all yes → coordinator logs commit; coordinator crash leaves participants in doubt holding locks.	9
24	Kafka log vs RabbitMQ queue?	Log: ordered replayable partitions, offsets; queue: per-message ack and load balancing, no replay.	11
25	Keep DB, index and cache in sync; make payments exactly-once?	No dual writes: CDC/event log with idempotent consumers; end-to-end operation ID with a unique constraint.	11, 12

0. Preface — how to read the book

The lasting principles and trade-offs underneath the buzzwords (NoSQL, sharding, ACID, CAP, MapReduce, real-time).

Must-know

- **Data-intensive** app — the hard part is the data (amount, complexity, speed of change); **compute-intensive** — the bottleneck is CPU. The book is about the first kind.
- **Forces behind the change** — internet-scale companies, business agility, open source winning, many cores + faster networks (parallelism), cheap distribution via IaaS, less tolerance for downtime.
- **Central promise** — tools change fast; principles and trade-offs change slowly. Learn those and you can place any new tool.
- **Scope** — architecture of data systems and their trade-offs; not install guides, not ops/security; "Big Data" is avoided as too vague. Sharper contrasts are used instead: **single-node vs distributed**, **online/interactive vs offline/batch**.
- **"You're not Google"** — building for scale you do not need is premature optimisation, but picking the right tool still matters; relational is important but not the last word.

Key facts & numbers

Part	Chapters	Theme
I · Foundations	1–4	Reliability/scalability/maintainability, data models, storage engines, encoding
II · Distributed Data	5–9	Replication, partitioning, transactions, distributed-systems trouble, consistency & consensus
III · Derived Data	10–12	Batch, stream, the future (integration, unbundling, correctness, ethics)

In the interview

Q: What question should you ask of any new data tool? **A:** What data model does it use, what guarantees does it give, and what does it give up to get them?

TL;DR

Part I is the prerequisite for Part II, Part II for Part III. Judge tools by their trade-offs, not their marketing.

1. Reliable, Scalable, and Maintainable Applications

The three nonfunctional goals every data system is judged by, and how to describe load and performance with numbers.

Must-know

- **Data system** — umbrella for databases, caches, search indexes, queues, batch/stream processors; the tools overlap (Redis as a queue, Kafka with database-like durability) and apps glue several together into a composite system.
- **Functional vs nonfunctional** — what the system does vs general properties it must have (reliability, scalability, security, maintainability...).
- **Reliability** — keeps working correctly, at the needed performance, even when things go wrong.
- **Fault vs failure** — a **fault** is one component deviating from spec; a **failure** is the whole system no longer serving users. Goal: stop faults turning into failures (**fault tolerance**).
- **Hardware faults** — mostly random and independent; handled by redundancy (RAID, dual power, hot-swap CPUs) and increasingly by software tolerating whole-machine loss (enables **rolling upgrades**).
- **Software errors** — **systematic**, correlated across nodes (same bug everywhere at once); cause **cascading failures**. Mitigate with testing, process isolation, restarts, measuring, monitoring.
- **Human errors** — the leading cause of outages. Mitigate with good APIs/abstractions, sandboxes, thorough testing, quick rollback and gradual rollout, recompute tools, **telemetry**, training.
- **Chaos Monkey** — Netflix tool that kills processes on purpose so the fault-handling path stays exercised.
- **Scalability** — not a yes/no label; it is "what are our options if load grows like X?"
- **Load parameters** — the few numbers that describe load (requests/s, read/write ratio, active users, cache hit rate, **fan-out**).
- **Throughput vs response time** — records/s (batch) vs time from request to reply as the client sees it (online).
- **Response time vs latency** — response time = service time + network + queueing delays; **latency** is only the time spent waiting to be handled.
- **Percentiles** — p50 (median) = typical experience; **p95/p99/p999** = the **tail**. Use them, not the mean.
- **Head-of-line blocking** — a few slow requests in front of a queue delay fast ones behind them → measure on the **client side**.
- **Tail latency amplification** — one user request that needs many backend calls is only as fast as the slowest call.
- **Scale up vs out** — bigger machine (vertical) vs many machines (**shared-nothing**, horizontal). Stateless services scale out easily; stateful data is harder.
- **Elastic** — adds resources automatically when load rises; manual scaling is simpler with fewer surprises.
- **Maintainability = operability + simplicity + evolvability.**
- **Accidental complexity** — complexity from the implementation, not the problem; removed with good **abstractions**. A project drowning in it is a **big ball of mud**.

Key facts & numbers

Fact	Value
Hard disk MTTF	~10–50 years → a 10,000-disk cluster loses ~1 disk/day
Outages involving hardware (study cited)	only 10–25%; operator config errors lead
Systematic bug example	leap second, 30 June 2012, Linux kernel bug hung many apps at once
Twitter post tweet	4.6k req/s avg (12k peak)
Twitter home timeline read	300k req/s (~100× writes)
Avg followers per tweet	~75 → ~345k timeline-cache writes/s
Amazon percentile target	p999 (not p9999: too costly)
Amazon cost of latency	+100 ms → −1% sales; others: +1 s → −16% satisfaction
Example SLA	median < 200 ms, p99 < 1 s, met ≥ 99.9% of the time
Cheap percentile algorithms	forward decay, t-digest, HdrHistogram

Twitter timeline	Approach 1: fan-out on read	Approach 2: fan-out on write	Hybrid (final)
Work at post time	insert one row	push tweet into every follower's cached timeline	write fan-out for normal users
Work at read time	join follows + tweets, merge, sort	read one cache	merge in celebrity tweets at read time
Breaks when	reads are 100× writes	a user has 30M+ followers	—

Maintainability part	Means	How
Operability	easy for ops to keep it running	monitoring, automation, no single-machine dependency, docs, sane defaults, self-healing with manual override, predictable behaviour
Simplicity	easy for new engineers to understand	remove accidental complexity with abstractions
Evolvability	easy to change for new requirements	simple, well-abstracted systems are easier to adapt

In the interview

Q: Fault vs failure? **A:** A fault is one part going wrong; a failure is the whole system stopping service. You cannot prevent all faults, so design fault tolerance and test it (e.g. Chaos Monkey).

Q: Why are software bugs worse than hardware faults? **A:** Hardware faults are mostly independent; a bug is systematic and hits every node with the same input at once (2012 leap second), so it can take down the whole fleet.

Q: Design the Twitter home timeline — fan-out on write or read? **A:** Reads outnumber posts ~100×, so precompute timelines on write; but celebrities make write fan-out explode, so use a hybrid that fetches celebrity tweets at read time. The key load parameter is the distribution of followers per user.

Q: Why percentiles, not the average? **A:** The mean hides how many users suffer; percentiles show the distribution, and the tail often belongs to the most valuable users and is amplified by fan-out.

Q: Can you average p99s from 10 servers? **A:** No — merge the histograms instead.

Q: Where should latency be measured? **A:** On the client, because queueing and head-of-line blocking add delay the server never sees.

Q: Scale up or out? **A:** Up is simpler but expensive at the high end; out goes further but adds complexity for stateful data. Real systems mix both, and the architecture is usually redesigned at each order-of-magnitude of growth.

Q: What makes a system maintainable? **A:** Operability, simplicity (no accidental complexity) and evolvability.

Q: When is it OK to sacrifice reliability? **A:** Deliberately, to cut development cost (a prototype for an unproven market) or operating cost (thin margins) — as long as you know you are cutting corners.

Gotchas

- "The system is scalable" → meaningless; say which load parameter grows and what your options are.
- Load test that waits for each response before sending the next → keeps queues short and makes response times look better than reality.
- Latency ≠ response time.
- Same bytes/s ≠ same architecture: 100,000 × 1 kB req/s and 3 × 2 GB req/min need very different designs — there is no generic scalable design.
- Optimising p9999 → usually not worth it; it is dominated by random events you cannot control.

TL;DR

Reliability = tolerate hardware, software and human faults; scalability = describe load and percentiles with numbers, then pick options; maintainability = operability, simplicity, evolvability.

2. Data Models and Query Languages

Relational vs document vs graph models, and declarative vs imperative ways to query them.

Must-know

- **Relational model** (Codd, 1970) — relations (tables) of unordered tuples (rows); won because the **query optimizer** picks access paths automatically.
- **NoSQL drivers** — scale (big data, high write throughput), open source, specialised queries, frustration with rigid schemas. Leads to **polyglot persistence**.
- **Impedance mismatch** — awkward translation between app objects and rows; ORMs (ActiveRecord, Hibernate) reduce boilerplate but do not remove it.
- **One-to-many** = a tree (fits documents); **many-to-one / many-to-many** = needs joins (fits relational/graph).
- **Normalization** — store each human-meaningful fact once, reference it by **ID**; IDs never need to change. **Denormalization** duplicates data to avoid joins at the cost of keeping copies in sync.
- **Hierarchical model (IMS)** — one big tree like JSON; bad at many-to-many; no joins. **Network model (CODASYL)** — records with many parents; programmer follows **access paths** by hand.
- **Document databases** repeat IMS for one-to-many, but use **document references** (like foreign keys) resolved by joins or follow-up queries for many-to-one.
- **Schema-on-read vs schema-on-write** — "schemaless" is misleading; the structure is implicit and interpreted by the reader (like dynamic typing) vs enforced on write (like static typing).
- **Locality** — a document is one contiguous string (JSON/XML/BSON), so loading it whole is fast; but the whole thing is loaded and updates often rewrite it → keep documents small. Relational

locality features: Spanner interleaved tables, Oracle multi-table index clusters, Bigtable column families.

- **Convergence** — relational DBs added JSON/XML support; RethinkDB supports joins; some MongoDB drivers resolve references (a slower **client-side join**).
- **Declarative** (SQL, CSS) — say *what* result you want; the optimizer decides *how*; no promised order, so storage can change; easy to parallelise. **Imperative** — you give the operations and their order.
- **MapReduce** — between declarative and imperative: you write `map` and `reduce` as **pure functions** so they can run anywhere, in any order, and be retried. MongoDB added the declarative **aggregation pipeline** (2.2).
- **Property graph** — vertices and edges each with an ID and key-value properties; edges have tail, head and a label. Any vertex can connect to any other; traverse forwards and backwards.
- **Triple-store** — (subject, predicate, object); object is either a value (a property) or another vertex (an edge). RDF/Turtle, queried with **SPARQL**.
- **Datalog** — rules build derived predicates from facts; a subset of Prolog; the foundation later languages build on (Datomic, Cascalog).

Key facts & numbers

	Document	Relational	Graph
Best for	self-contained trees, loaded whole	many-to-one, many-to-many via joins	anything related to everything; variable-length traversals
Schema	usually on read	on write	usually not enforced
Joins	weak (RethinkDB yes; MongoDB no; CouchDB only predeclared views)	strong	traversals are the point
Query language	MongoDB aggregation pipeline, MapReduce	SQL	Cypher (Neo4j), SPARQL, Datalog
Examples	MongoDB, RethinkDB, CouchDB, Espresso	—	Neo4j, Titan, InfiniteGraph (property); Datomic, AllegroGraph (triples)

Graph DB vs CODASYL	Graph database	CODASYL network model
Schema on links	none; any vertex to any vertex	schema restricts which record types nest
Reaching a record	by ID or index	only by following access paths
Order	unordered vertices/edges	ordered child sets
Query style	declarative (Cypher, SPARQL)	imperative only

Fact	Value
Recursive query in SQL	`WITH RECURSIVE` (SQL:1999) — book's example 29 lines vs 4 in Cypher
MySQL `ALTER TABLE`	copies the whole table → possible long downtime; workaround: nullable column, fill at read time

Code to remember

```
MATCH
  (p) -[:BORN_IN]->  () -[:WITHIN*0..]-> (:Location {name:'India'}),
  (p) -[:LIVES_IN]-> () -[:WITHIN*0..]-> (:Location {name:'Canada'})
RETURN p.name
```

`*0..` = follow WITHIN edges zero or more times; the optimizer decides forward or backward traversal.

In the interview

Q: Relational or document? **A:** A self-contained one-to-many tree loaded whole → document (locality, schema flexibility, no shredding). Many-to-one/many-to-many → relational joins. Highly interconnected → graph.

Q: Is a document DB schemaless? **A:** No — it is schema-on-read; the code still assumes a structure, the database just does not enforce it.

Q: Why store IDs rather than text? **A:** Normalization: one fact stored once, renames are one update, no inconsistent copies; costs joins.

Q: Why did relational beat CODASYL? **A:** CODASYL made programmers navigate access paths by hand; relational let the optimizer choose paths, so new indexes and queries did not require rewriting code.

Q: Why does declarative matter? **A:** Shorter queries, the engine can change storage or algorithms without breaking them, and parallel execution is much easier.

Q: Where does MapReduce fit? **A:** Between: snippets of code (pure functions) run by the framework; harder to coordinate and gives less room to optimise than a declarative pipeline.

Q: Graph queries in SQL? **A:** Possible with recursive CTEs, but clumsy compared with Cypher.

Q: Property graph vs triple-store? **A:** Mostly equivalent ways of saying the same thing.

Gotchas

- "Document DBs are schemaless" → schema-on-read.
- Big documents → whole doc loaded and rewritten on growth; keep them small.
- Client-side joins → extra round-trips, less optimisation than in-database joins.
- Emulating one model in another is possible but awkward — pick the model that fits.
- Map and reduce with side effects or extra queries → breaks retries and parallelism.

TL;DR

Documents for trees, relational for joins, graphs for everything-connects-to-everything; declarative languages let the engine optimise for you.

3. Storage and Retrieval

How storage engines lay data out: logs, hash indexes, LSM-trees, B-trees, and column stores for analytics.

Must-know

- **Log** — append-only file: cheapest possible write; reading without an index is O(n).
- **Index** — extra structure derived from the data; speeds reads, slows every write → index deliberately.
- **Hash index (Bitcask)** — in-memory map key → byte offset in the log. Great for many updates to few keys (play count per video URL). Keys must fit in RAM; no efficient range queries.
- **Segments + compaction** — close a segment at a size limit; compaction throws away duplicate keys, merging combines segments; old segments are deleted. **Tombstone** = delete marker.
- **SSTable** — segment with keys **sorted**; merge like mergesort; a **sparse** in-memory index is enough; blocks between index entries can be compressed.
- **LSM-tree** — write → append to log (crash safety) + insert into in-memory balanced tree (**memtable**); at a few MB flush to an SSTable; reads check memtable then newest→oldest SSTables; background

merge/compaction. Used by LevelDB, RocksDB, Cassandra, HBase, Lucene (term dictionary).

- **Bloom filter** — tells an LSM "key definitely not in this SSTable", avoiding reads for missing keys.
- **Size-tiered vs leveled compaction** — merge newer/smaller into older/larger (HBase) vs split key range into small tables across levels, more incremental and less disk (LevelDB, RocksDB); Cassandra supports both.
- **B-tree** — fixed-size **pages** (traditionally 4 KB) overwritten in place; a root, internal pages holding key-range references, leaf pages; **branching factor** usually several hundred; full page → **split** and update parent → always balanced, O(log n) depth, usually 3–4 levels.
- **WAL (redo log)** — append-only log of every B-tree change before applying it, so a crash mid-split can be repaired. **Latches** (light locks) protect concurrent access. Alternative: **copy-on-write** pages (LMDB).
- **B-tree optimisations** — abbreviated keys in internal pages (B+ tree flavour), leaf pages laid out in order, sibling pointers, fractal trees.
- **Write amplification** — one logical write → several physical writes (WAL + page; repeated compaction). Limits throughput and wears out SSDs.
- **Secondary index** — keys not unique: store a list of row IDs or make keys unique by appending the row ID.
- **Heap file vs clustered vs covering index** — pointer to a row stored elsewhere vs full row in the index (InnoDB primary key; SQL Server one per table) vs some columns included.
- **Concatenated index** — several fields joined in one key (phone book: last, first); useless for "first name only". **Multi-dimensional** (R-tree, PostGIS; HyperDex) for geo-rectangles and similar.
- **Fuzzy index** — Lucene's in-memory automaton over terms supports search within an edit distance.
- **In-memory DBs** (VoltDB, MemSQL, RAMCloud, Redis) — faster because they avoid **encoding** data for disk, not because they avoid disk reads (the OS cache already does that). Can also offer data models hard on disk (Redis sets, queues). **Anti-caching** evicts to disk.
- **OLTP vs OLAP** — see table; analytics goes to a separate **data warehouse** loaded by **ETL** (extract–transform–load).
- **Star schema** — central **fact table** (one row per event) with foreign keys to **dimension tables** (who, what, where, when, how, why). **Snowflake** normalises dimensions further.
- **Column-oriented storage** — each column in its own file, same row order in every file; queries read only the columns they need.
- **Column compression** — bitmap encoding + **run-length encoding**; fits CPU cache → **vectorised processing**.
- **Sort order** — sort whole rows (never columns independently) by a chosen key; first sort key compresses best. C-Store/Vertica keep **several sort orders** on different replicas.
- **Writes to column stores** — buffer in memory (LSM-style), merge into column files in bulk; queries combine both (Vertica).
- **Materialized view / data cube** — precomputed aggregates stored on disk: fast reads, costlier writes, less flexible than raw data.

Key facts & numbers

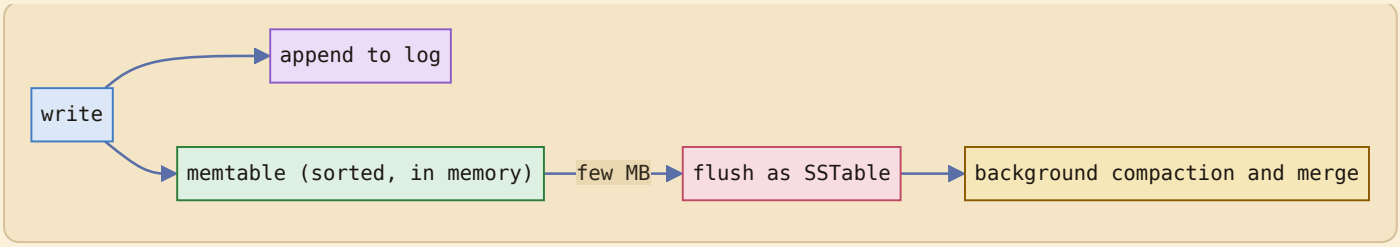
Aspect	B-tree	LSM-tree
Typical strength	reads, predictable latency	writes, throughput
Write pattern	random page overwrites	sequential segment writes
Write amplification	WAL + full page writes (sometimes twice)	repeated compaction (often lower overall)
Space	fragmentation leaves empty space	more compact, better compression
Background work	none	compaction can hurt tail latency / fall behind
Key location	exactly one place (good for range locks in transactions)	possibly many segments
Maturity	very mature; default in relational DBs	increasingly popular in new datastores

Storage family	Rule	Members
Log-structured	only append; delete obsolete files; never modify	Bitcask, SSTables, LSM-trees, LevelDB, RocksDB, HBase, Cassandra, Lucene
Update-in-place	overwrite fixed-size pages	B-trees (all major relational DBs, many others)

Property	OLTP	OLAP
Read pattern	few records by key	aggregate over very many records
Write pattern	low-latency random writes from users	bulk import (ETL) or event stream
Used by	end users via the app	internal analysts
Data represents	latest state	history of events
Size	GB–TB	TB–PB
Bottleneck	disk seek time	disk bandwidth

Index option	Stores	Read cost	Write/storage cost
Heap (nonclustered)	pointer to row	extra hop	lowest
Covering	some columns	no hop for covered queries	medium
Clustered	entire row	no hop	highest

Fact	Value
4 levels × branching 500 × 4 KB pages	up to 256 TB
Typical B-tree depth	3–4 levels
Memtable flush threshold	a few MB
Virtual vs materialized view	query expanded on the fly vs results stored on disk and kept updated



In the interview

Q: B-tree or LSM-tree? **A:** B-tree: in-place pages + WAL, each key in one place, fast predictable reads — default for relational DBs. LSM: memtable + sequential SSTables + compaction, higher write throughput and compression but reads may check several files and compaction can hurt tail latency. Write-heavy → LSM; read-heavy/transactional → B-tree; benchmark your workload.

Q: How does an LSM survive a crash? **A:** Every write is also appended to a log; on restart the log is replayed into the memtable; the log is discarded once its memtable is flushed.

Q: How does an LSM avoid reading every SSTable for a missing key? **A:** A Bloom filter per SSTable.

Q: What if compaction cannot keep up? **A:** Unmerged segments pile up, reads slow down, disk fills; engines usually do not throttle writes, so monitor it.

Q: How does a B-tree stay balanced and crash-safe? **A:** Overflowing pages split and the parent gets a new boundary (O(log n) depth); a WAL makes multi-page changes recoverable; latches handle concurrency.

Q: Why a separate data warehouse? **A:** Analyst scans would hurt user-facing OLTP latency and OLTP indexes are poor for scans, so ETL copies data into a read-optimised (often columnar, star-schema) warehouse.

Q: Why are column stores fast for analytics? **A:** Queries touch a few of 100+ columns, so read only those; columns compress well (bitmaps, RLE), chunks fit CPU cache for vectorised processing.

Q: Are Cassandra/HBase column stores? **A:** No — column families are still row-oriented within each family and do not use column compression.

Q: Why are in-memory DBs faster? **A:** They skip encoding in-memory structures into disk format.

Q: Why not a (lat, lon) concatenated index for "shops in this rectangle"? **A:** It narrows latitude but scans all longitudes; use a spatial index (R-tree) or a space-filling curve.

Gotchas

- Sorting each column independently → breaks row reconstruction; sort whole rows.
- "In-memory = no disk" → most write a log/snapshots to disk for durability; the win is avoiding encoding.
- Data cube only → cannot answer questions along dimensions you did not include; keep raw data too.
- Materialized views in OLTP → every write must update them; they fit read-heavy warehouses.
- "LSM always beats B-tree for writes" → depends on workload and configuration; test it.

TL;DR

Logs and LSM-trees turn random writes into sequential ones; B-trees give each key one home and predictable reads; analytics wants columns, compression and sort order.

4. Encoding and Evolution

Turning data into bytes (JSON, Thrift, Protocol Buffers, Avro) and changing schemas safely while old and new code run side by side.

Must-know

- **Encoding (serialization/marshalling)** — in-memory objects → self-contained bytes; **decoding** is the reverse.
- **Backward compatibility** — new code reads old data (easy: you know the old format). **Forward compatibility** — old code reads new data (harder: old code must ignore additions).
- **Why both** — **rolling upgrades** and un-updated clients mean old and new versions coexist and read each other's data.
- **Language-specific formats** (Java Serializable, Ruby Marshal, Python pickle, Kryo) — language lock-in, remote-code-execution risk when decoding untrusted bytes, weak versioning, poor efficiency. Only for very short-lived data.
- **JSON/XML/CSV** — universal and readable, but number ambiguity (integers > 2^53 not exact in JavaScript doubles), no binary strings (Base64, +33%), optional/complex schemas, CSV has none.
- **Binary JSON** (MessagePack, BSON...) — still stores field names; saves little.
- **Thrift / Protocol Buffers** — schema in an IDL; each field has a **tag number**; data holds tag + type + value, never names. Code generation for many languages.
- **Tag rules** — rename freely; **never change or reuse a tag**; new fields must be optional or have a default; only optional fields may be removed.
- **Avro** — no tags, no type markers; values in schema order. Decoding needs the exact **writer's schema**, resolved against the **reader's schema** by **field name**: unknown fields ignored, missing ones filled with defaults.
- **Avro rules** — add/remove only fields that have a **default**; null via a **union** (`union { null, long }`); changing a type works if Avro can convert it.
- **Where the writer's schema comes from** — object container file header (once per file), a per-record **version number** + schema list in the DB (Espresso), or negotiation at connection setup (Avro RPC).
- **Dynamically generated schemas** — Avro suits schemas generated from DB tables (no tag bookkeeping).
- **Merits of schemas** — compact, always-up-to-date documentation, compatibility checks before deploy, code generation in static languages (ASN.1 did it first, 1984).
- **Data outlives code** — a DB holds values written years apart; preserve **unknown fields** on read-modify-write; archive dumps with the latest schema (Avro container files, Parquet).
- **REST vs SOAP vs RPC** — REST: philosophy on HTTP, simple formats; SOAP: XML protocol with WSDL, heavy tooling; RPC: tries to make remote calls look local (**location transparency**) — flawed.
- **Service compatibility** — assume servers upgrade first: **backward-compatible requests, forward-compatible responses**. Version via URL, `Accept` header, or per-API-key settings.
- **Message broker** — buffers, redelivers, hides addresses, fans out, decouples; one-way and async (replies need a separate queue). Consumers that republish must keep fields they do not understand.
- **Distributed actors** (Akka, Orleans, Erlang OTP) — location transparency fits better because actors already assume message loss; still need compatible encodings for rolling upgrades.

Key facts & numbers

Encoding (book's example record)	Bytes
JSON, whitespace removed	81
MessagePack	66
Thrift BinaryProtocol	59
Thrift CompactProtocol	34
Protocol Buffers	33
Avro	32

Family	Examples	Readable	Schema	Compatibility
Language-specific	Java Serializable, pickle, Kryo	No	implicit in classes	often none; insecure
Textual	JSON, XML, CSV	Yes	optional / none	depends on usage
Binary JSON	MessagePack, BSON	No	none	like JSON
Tag-based binary	Thrift, Protocol Buffers	No	required, with tags	clear rules via tags
Name-resolved binary	Avro	No	writer's + reader's	name matching + defaults

Change	Protobuf/Thrift	Avro
Rename field	OK (names not encoded)	only via aliases (names matter)
Add field	must be optional/defaulted, new tag	must have a default
Remove field	only optional; never reuse its tag	only if it had a default
int32 → int64	new code fine; old code may truncate	OK if convertible
optional → repeated	protobuf OK (old code sees last element)	—

Dataflow mode	Who encodes / decodes	Compatibility needs
Database	writer encodes, (later) reader decodes	both directions; keep unknown fields
REST/RPC	client encodes request, server decodes and replies	backward requests, forward responses
Async messages	sender encodes, recipient decodes	both; preserve unknown fields on republish

Local call	Network call
predictable	may be lost, slow, or target down
returns / throws	may **time out** with unknown outcome
retry is your choice	retry may duplicate → need **idempotence**
steady latency	ms to many seconds
pass pointers	encode every parameter to bytes

Code to remember

```
message Book {
  required string book_title = 1;   // tag 1 is forever
  optional int64  page_count = 2;
  repeated string genres     = 3;   // a list
}
```

In the interview

Q: Backward vs forward compatibility — why both? **A:** New reads old vs old reads new; during rolling upgrades both versions run at once and read each other's data.

Q: How does protobuf evolve schemas? **A:** Fields are identified by numeric tags; old code skips unknown tags (the type gives the length), new code reads old data if new fields are

optional/defaulted. Never change or reuse tags.

Q: How is Avro different? **A:** No tags; the reader resolves the writer's schema against its own by field name, with defaults for missing fields. Great for auto-generated schemas.

Q: How does an Avro reader find the writer's schema? **A:** File header, per-record version number + schema registry, or connection negotiation.

Q: Which compatibility does a database need? **A:** Both, plus preserving unknown fields when old code rewrites a record.

Q: Why is RPC's "remote looks local" flawed? **A:** Network calls can be lost, time out with unknown outcome, be duplicated on retry, vary wildly in latency, and cross type systems.

Q: REST or RPC? **A:** REST for public APIs (debuggable, universal tooling); binary RPC (gRPC, Thrift) inside an organisation where performance matters.

Q: Broker vs direct RPC? **A:** Broker buffers, redelivers, decouples and fans out; costs you a one-way asynchronous flow.

Q: Why avoid pickle/Java serialization for stored data? **A:** Language lock-in, security holes, poor versioning and efficiency.

Gotchas

- Old code reads a record, updates it, writes it back → new field silently lost; preserve unknown fields.
- Adding a `required` protobuf field or an Avro field without a default → breaks reading old data.
- 64-bit IDs as JSON numbers → JavaScript loses precision; Twitter sends IDs as strings too.
- Binary JSON "saves space" → barely (66 vs 81 bytes); schema formats get 32–34.
- Reusing a deleted field's tag → old data decoded as the wrong field.

TL;DR

Code and data of different ages coexist, so every encoding must be backward and forward compatible; schema-driven binary formats (tags or writer/reader schemas) make that precise and compact.

5. Replication

Keeping a copy of the same data on several machines: single-leader, multi-leader and leaderless, and what replication lag does to users.

Must-know

- **Why distribute data (Part II intro)** — scalability, fault tolerance/high availability, latency (servers near users). Architectures: **shared-memory** (scale up), **shared-disk** (NAS/SAN, locking limits it), **shared-nothing** (nodes coordinate in software over a normal network — the focus of Part II).
- **Replication vs partitioning** — same data on many nodes vs cutting data into pieces; independent, usually combined.
- **Leader-based (master–slave, active/passive)** — clients write to the leader; it sends a **replication log** to followers; reads can go to any replica.
- **Synchronous vs asynchronous** — wait for a follower's ack (follower guaranteed current, but one stalled follower blocks all writes) vs don't wait (fast, but confirmed writes can be lost). **Semi-synchronous** = one sync follower, rest async. Fully async is common, weakening durability.
- **New follower** — take a consistent snapshot tied to a log position (LSN / binlog coordinates), copy it, then apply all changes since that position.

- **Follower failure** — catch-up recovery from its log. **Leader failure** — **failover**: detect (timeout, e.g. 30 s), choose new leader (freshest replica; a consensus problem), reconfigure clients; old leader must step down.
- **Failover dangers** — lost unreplicated writes (durability broken), clashes with outside systems (GitHub reused MySQL autoincrement keys also used in Redis → private data shown to wrong users), **split brain**, timeout choice (too short → needless failovers that worsen overload). Some teams fail over manually.
- **Fencing / STONITH** — shut one leader down when two are detected; badly designed versions shut down both.
- **Replication lag** — async followers can be behind; **eventual consistency** is vague about how far.
- **Read-after-write (read-your-writes)** — users always see their own submitted data.
- **Monotonic reads** — once you have seen newer data, you never see older data later.
- **Consistent prefix reads** — writes seen in the order they happened (question before answer); mainly a partitioned-store problem.
- **Multi-leader** — several nodes accept writes and replicate to each other asynchronously. Use cases: **multi-datacenter**, **offline clients** (each device is a leader; CouchDB), **collaborative editing**.
- **Conflict avoidance** — route all writes for a record to one home leader; breaks when that leader changes.
- **Convergent resolution** — all replicas must end at the same value: highest unique ID / **LWW** (data loss), higher replica ID wins (data loss), merge values, or record the conflict for the app.
- **Custom resolution** — **on write** (Bucardo Perl handler, background, cannot ask user) or **on read** (CouchDB returns all versions). Resolution is per row/document, not per transaction.
- **Automatic resolution** — **CRDTs** (Riak 2.0), mergeable persistent data structures (Git-like three-way merge), **operational transformation** (Etherpad, Google Docs). Amazon's cart once kept additions but not removals → deleted items came back.
- **Topologies** — circular (MySQL default), star/tree, all-to-all. Loops avoided by tagging writes with node IDs. Circular/star: one failed node blocks replication. All-to-all: messages can overtake each other → causality problem; timestamps are not enough → **version vectors**.
- **Leaderless (Dynamo-style)** — Riak, Cassandra, Voldemort: client (or coordinator) writes to and reads from several replicas in parallel; no failover.
- **Read repair vs anti-entropy** — fix stale replica noticed during a read (good for hot data) vs background process copying missing data in no order (Voldemort has none).
- **Quorum** — n replicas, write needs w acks, read asks r; **w + r > n** → read and write sets overlap. Typical n = 3 or 5, w = r = ⌈(n+1)/2⌉.
- **Sloppy quorum + hinted handoff** — when home nodes are unreachable, accept writes on other nodes and hand them back later; better write availability, but no overlap guarantee. On by default in Riak; off in Cassandra and Voldemort.
- **Concurrent** — neither operation knew about the other (not about wall-clock time). **Happens-before** — B knows about, depends on, or builds on A.
- **Version number algorithm** — server keeps a version per key; client must read before write and send back the version it read; server overwrites values ≤ that version and keeps higher ones as **siblings**; client merges. Across replicas → **version vector** (Riak: dotted version vectors, "causal context").
- **Tombstone** — marker kept in place of a deleted item so the deletion survives sibling merges.

Key facts & numbers

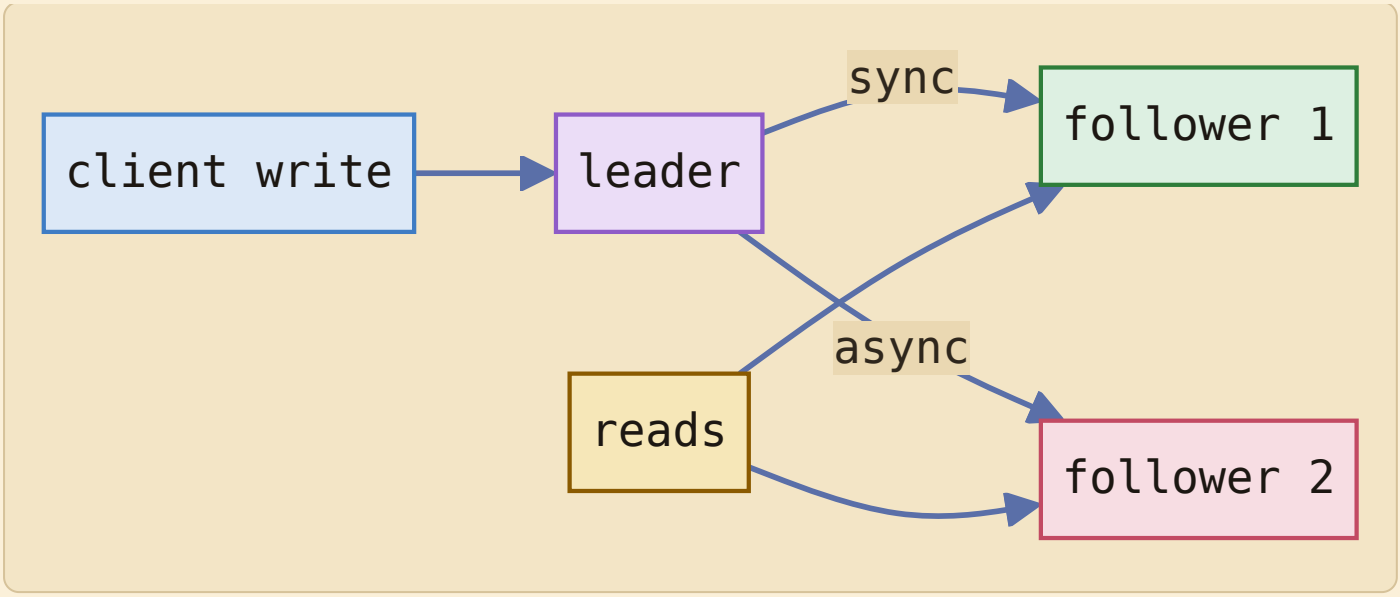
Approach	How writes flow	Pros	Cons
Single-leader	all writes via one leader → followers	simple; no conflicts	writes depend on one leader; failover is fraught
Multi-leader	each write to one of several leaders; leaders sync async	tolerates node/network trouble, local write latency	conflicts; hard to reason about
Leaderless	write/read several nodes; quorums	no failover; very available	very weak consistency; conflicts

Replication log method	Sends	Pros	Cons	Examples
Statement-based	SQL statements	compact	nondeterminism (NOW(), RAND()), ordering, side effects	MySQL < 5.1, VoltDB
WAL shipping	disk-block changes	exact copy	tied to storage format; upgrades need downtime	PostgreSQL, Oracle
Logical (row-based)	row changes	version-independent; good for CDC	a second log format	MySQL binlog (row mode)
Trigger-based	whatever app code records	flexible	overhead, bugs	Databus, Bucardo

Anomaly	User sees	Guarantee	Fix
Own write missing	"my post disappeared"	read-after-write	read own data from leader (or for 1 min after a write); track last-write timestamp/LSN
Time goes backwards	item appears then vanishes	monotonic reads	pin each user to one replica (hash of user ID)
Effect before cause	answer before question	consistent prefix	causally related writes in one partition; track causality

Quorum config	Tolerates
n = 3, w = r = 2	1 unavailable node
n = 5, w = r = 3	2 unavailable nodes
w = n, r = 1	fast reads; one node down → all writes fail

Multi-DC	Single-leader	Multi-leader
Write latency	every write crosses to the leader's DC	local writes, async sync
DC outage	failover to another DC	each DC keeps running
Network trouble	very sensitive (sync over internet)	tolerates interruptions



In the interview

Q: Single-leader vs multi-leader vs leaderless? **A:** Single: one write node, ordered log, no conflicts, fragile failover. Multi: a leader per DC/device, local writes, must resolve conflicts. Leaderless: quorum reads/writes, no failover, weak consistency.

Q: Why not multi-leader inside one datacenter? **A:** Conflicts and trouble with autoincrement keys, triggers and constraints rarely pay off on a fast, reliable local network.

Q: Is AWS DynamoDB leaderless? **A:** The book notes it uses single-leader replication; Dynamo was Amazon's internal system.

Q: Sync vs async? **A:** Sync guarantees a current copy but blocks writes if the follower stalls; async is fast but loses confirmed writes on leader failure; semi-sync is the compromise.

Q: What can go wrong in failover? **A:** Lost writes, reused keys clashing with other systems, split brain, and bad timeout choice.

Q: Explain quorums. **A:** w + r > n makes read and write sets overlap, so at least one replica read has the newest value; read repair and anti-entropy fix stragglers.

Q: Does w + r > n guarantee fresh reads? **A:** No — sloppy quorums, LWW with clock skew, writes concurrent with reads, partially failed writes not rolled back, restores from stale replicas, timing edge cases. It is a probability knob, not linearizability.

Q: Fix "my comment disappeared after reload"? **A:** Read-after-write: read own content from the leader or require replicas that have caught up to the user's last write position.

Q: How do you detect concurrent writes, and why not LWW? **A:** Version numbers/vectors show whether one write knew of the other; keep concurrent ones as siblings to merge. LWW silently drops writes — safe only for write-once keys (e.g. UUID keys in Cassandra).

Q: Offline-first notes app syncing phone and laptop? **A:** Multi-leader with very long lag; plan conflict resolution (CRDTs or on-read like CouchDB).

Gotchas

- "Eventual consistency" → says nothing about how long; lag can be seconds or minutes under load.
- Timestamps to order multi-leader writes → clocks drift; use version vectors.
- Making all followers synchronous → any one outage halts all writes.
- Multi-device read-your-writes → last-write metadata must be central; route all of a user's devices to one DC.
- Merging siblings by union → deletions reappear; use tombstones or CRDTs.

- PostgreSQL BDR gives no causal ordering; Tungsten Replicator for MySQL does not detect conflicts — read the docs and test.

TL;DR

Replication buys availability, latency and read scale; async lag creates read-your-writes, monotonic and prefix anomalies; multi-leader and leaderless trade simplicity for conflicts you must detect (version vectors) and resolve.

6. Partitioning

Splitting a dataset too big for one machine into partitions (shards), indexing them, rebalancing them, and routing requests to them.

Must-know

- **Partition** — each record belongs to exactly one partition; a.k.a. shard (MongoDB, Elasticsearch, SolrCloud), region (HBase), tablet (Bigtable), vnode (Cassandra, Riak), vBucket (Couchbase).
- **Partitioning + replication** — each partition is replicated; a node can be leader for some partitions and follower for others.
- **Goal** — spread data and query load evenly. **Skew** = unfair split; **hot spot** = one partition with disproportionate load.
- **Random assignment** — even load, but you would have to query every node to find a record.
- **Key-range partitioning** — each partition owns a contiguous range of sorted keys (like encyclopedia volumes); boundaries adapt to the data. Range scans are cheap; sequential keys (timestamps) create hot spots. Bigtable, HBase, RethinkDB, MongoDB before 2.4.
- **Hash partitioning** — each partition owns a range of hash values; spreads load, loses range queries. Needs a hash that is the same in every process (MD5 in Cassandra/MongoDB, Fowler–Noll–Vo in Voldemort) — not Java `hashCode()`.
- **Consistent hashing** — Karger et al.'s random-boundary method for CDN caches; "consistent" has nothing to do with replica or ACID consistency; works poorly for databases → say **hash partitioning**.
- **Compound key (Cassandra)** — first column hashed to pick the partition, remaining columns sort data within it: `(user_id, update_timestamp)` → efficient range over one user's timestamps.
- **Hot keys (celebrities)** — hashing does not help (same key, same hash). Append a random two-digit number → 100 keys; reads fetch all 100 and combine; only for the few hot keys, and you must track which keys are split.
- **Document-partitioned (local) secondary index** — each partition indexes only its own documents; write touches one partition; read must **scatter/gather** to all (tail-latency amplification).
- **Term-partitioned (global) secondary index** — the index itself is partitioned by term; read hits one partition; a write may touch several index partitions and is usually **asynchronous**.
- **Rebalancing requirements** — load fair afterwards; DB keeps serving during it; move no more data than necessary.
- **Hash mod N** — don't: changing N moves most keys.
- **Fixed number of partitions** — many more partitions than nodes (e.g. 10 nodes, 1,000 partitions); a new node steals whole partitions; key→partition mapping never changes. Caps node count; count is hard to choose.
- **Dynamic partitioning** — split a partition when it grows too big (HBase default 10 GB), merge when small. Empty DB starts with one partition → **pre-splitting** (HBase, MongoDB).
- **Partitions proportional to nodes** — fixed number per node (Cassandra 256); a new node splits random existing partitions; closest to original consistent hashing.

- **Automatic vs manual rebalancing** — automatic + automatic failure detection can turn a slow node into a **cascading failure**; a human approving suggested plans (Couchbase, Riak, Voldemort) is slower but safer.
- **Request routing (service discovery)** — (1) any node forwards, (2) a routing tier, (3) partition-aware clients. Hard part: everyone agreeing on the partition map → often **ZooKeeper**; Cassandra/Riak use **gossip**. DNS is fine for node IPs.
- **MPP** — analytic databases split complex queries into stages executed in parallel across partitions.

Key facts & numbers

	Key-range	Hash
Partition owns	a sorted key range	a range of hash values
Range queries	efficient	inefficient (order lost; MongoDB hash mode sends them to all partitions)
Hot-spot risk	high for sequential keys	lower
Typical rebalancing	dynamic split/merge	fixed partitions moved whole (dynamic also possible)
Users	Bigtable, HBase, RethinkDB, MongoDB < 2.4	Cassandra, MongoDB (MD5), Voldemort (FNV)

	Local (document-partitioned)	Global (term-partitioned)
Index lives	with each data partition	separately, partitioned by term
Writes	one partition	possibly several index partitions
Reads	scatter/gather to all	only partitions with the term
Freshness	updated with the doc	often async; may lag
Examples	MongoDB, Riak, Cassandra, Elasticsearch, SolrCloud, VoltDB	DynamoDB GSIs, Riak search, Oracle DW (either)

Rebalancing	# partitions	Partition size	Used by	Watch out
hash mod N	= nodes	varies	avoid	most keys move
Fixed number	fixed at setup	grows with data	Riak, Elasticsearch, Couchbase, Voldemort	choosing the count
Dynamic	grows with data	bounded (HBase 10 GB)	HBase, RethinkDB, MongoDB	starts with one; pre-split
Proportional to nodes	fixed per node	fairly stable	Cassandra (256/node), Ketama	random splits can be unfair

hash = 123456	N = 10	N = 11	N = 12
mod N → node	6	3	0

Routing	Who
ZooKeeper tracks assignment	HBase, SolrCloud, Kafka; Espresso via Helix
own config servers + mongos routers	MongoDB
gossip; any node forwards	Cassandra, Riak
routing tier moxi, no auto-rebalancing	Couchbase

In the interview

Q: Key-range vs hash partitioning? **A:** Range keeps order (cheap scans, hot-spot risk with sequential keys); hash spreads load evenly but kills range queries. Cassandra's compound key gets some of both.

Q: Timestamp-keyed sensor data overloads one node — fix? **A:** Prefix the key with the sensor name so writes spread; reading many sensors then needs one range query per sensor.

Q: Why is hash mod N bad? **A:** Changing N remaps most keys. Use a fixed large number of partitions, dynamic splitting, or partitions per node.

Q: Downside of a fixed partition count? **A:** Caps node count; too big → costly rebalancing/recovery, too small → overhead.

Q: Local vs global secondary index? **A:** Local: cheap writes, scatter/gather reads. Global: single-partition reads, multi-partition writes usually applied asynchronously (DynamoDB GSIs normally within a fraction of a second).

Q: Why not update global indexes synchronously? **A:** It needs a distributed transaction across index partitions, which many DBs lack.

Q: Celebrity hot key? **A:** Split it with a random suffix across ~100 keys, read and combine all; bookkeeping needed.

Q: How does a request find its partition? **A:** Forwarding node, routing tier, or partition-aware client; the partition map is kept consistent in ZooKeeper or spread by gossip.

Q: Should rebalancing be automatic? **A:** Risky: a merely slow node may be declared dead, and moving its load can cascade; keep a human in the loop.

Gotchas

- "Consistent hashing" in database docs → usually inaccurate; say hash partitioning.
- Java `hashCode()` for partitioning → differs across processes.
- Hashing a hot key → still one partition.
- Scatter/gather reads → tail latency grows with partition count.
- Dynamic partitioning on an empty DB → one node takes every write until the first split.
- Writes spanning partitions → partial failure is hard (later chapters).

TL;DR

Partition by key range (order, hot-spot risk) or hash (even load, no ranges); index locally (scatter reads) or globally (async multi-partition writes); rebalance by moving whole partitions, never mod N; route via ZooKeeper or gossip.

7. Transactions

What ACID really means, the weak isolation levels and the races they allow, and three ways to get serializability.

Must-know

- **Transaction** — groups several reads and writes into one logical unit that either commits or aborts; lets the app ignore many faults and concurrency issues and simply **retry** after an abort.
- **Atomicity** — on a fault, discard all of the transaction's writes; really **abortability**. Not the threading sense of "no one sees half-done state".
- **Consistency** — app-defined invariants hold; mostly the **app's** job (the DB checks only some, like foreign keys or uniqueness). "Consistency" also means replica consistency, consistent hashing and linearizability (CAP) elsewhere.
- **Isolation** — concurrent transactions don't interfere; the textbook ideal is **serializability**, but most DBs default to weaker levels.
- **Durability** — committed data is not lost: disk + WAL, or replication. Never perfect (correlated faults, SSD bugs, corruption).
- **BASE** — Basically Available, Soft state, Eventual consistency: vaguer than ACID; mainly "not ACID".
- **Single-object operations** — atomicity via a log, isolation via a per-object lock; plus atomic increment and compare-and-set. These are not "transactions" in the usual sense.
- **Why multi-object transactions** — foreign-key references, denormalized documents, secondary indexes must all stay in sync.
- **Read committed** — **no dirty reads** (only see committed data) and **no dirty writes** (only overwrite committed data). Implemented with row-level write locks; readers get the old committed value (no read locks, which would stall readers behind long writes).
- **Read skew (nonrepeatable read)** — seeing different parts of the DB at different points in time (Alice's $1,000 appears as $900). Hurts **backups** and **analytic queries / integrity checks**.
- **Snapshot isolation** — each transaction reads from a **consistent snapshot** taken at its start. Implemented with **MVCC**: rows tagged `created_by` / `deleted_by` txid; an update = delete + create; garbage collection removes invisible versions. **Readers never block writers, writers never block readers.**
- **Visibility rules** — ignore writes by transactions in progress at start, by aborted transactions, and by transactions with a later txid; everything else is visible.
- **Indexes under MVCC** — point to all versions and filter, or append-only/copy-on-write B-trees (CouchDB, Datomic, LMDB).
- **Lost update** — two concurrent read-modify-write cycles; one clobbers the other. Fixes: **atomic write** operations, **explicit locking** (`SELECT ... FOR UPDATE`), **automatic detection**, **compare-and-set**; with replication: commutative ops or sibling merge (LWW loses updates).
- **Write skew** — two transactions read the same data, then update **different** objects based on it, breaking an invariant (both on-call doctors go off call; double-booked room; duplicate username; double spending). A generalisation of lost update.
- **Phantom** — a write in one transaction changes the result of a search query in another. `FOR UPDATE` cannot lock rows that don't exist yet.
- **Materializing conflicts** — pre-create lock rows (e.g. room × 15-minute slots) to lock on; last resort, leaks concurrency control into the data model.
- **Actual serial execution** — one transaction at a time on one thread (VoltDB/H-Store, Redis, Datomic); feasible since ~2007 because RAM is cheap and OLTP transactions are short. Requires **stored procedures** (no interactive multi-round-trip transactions), data in memory, low write throughput per partition.
- **Two-phase locking (2PL)** — shared lock to read, exclusive to write, **held until commit/abort**. Writers block readers and writers; readers block writers. Deadlocks detected and one transaction aborted. **Predicate locks** / cheaper **index-range (next-key) locks** stop phantoms. MySQL InnoDB & SQL Server serializable, DB2 "repeatable read".
- **2PL ≠ 2PC** — 2PL is isolation on one DB; 2PC is atomic commit across nodes (Ch 9).

- **Serializable snapshot isolation (SSI)** — snapshot isolation + detecting serialization conflicts, abort at commit. **Optimistic**. Detects (1) **stale MVCC reads** (ignored an uncommitted write that later committed) and (2) **writes that affect prior reads** (non-blocking index-range "tripwires"). PostgreSQL serializable since 9.1; FoundationDB distributes detection.
- **Pessimistic vs optimistic** — wait if anything *might* go wrong (2PL, serial execution) vs carry on and check at commit (SSI); optimistic is faster with spare capacity and low contention, worse under high contention.

Key facts & numbers

Anomaly	What happens	Prevented by
Dirty read	read another's uncommitted write	read committed and stronger
Dirty write	overwrite another's uncommitted write	almost all implementations
Read skew	parts of DB seen at different moments	snapshot isolation (MVCC)
Lost update	concurrent read-modify-write; one lost	some SI implementations detect it; else `FOR UPDATE` / atomic ops
Write skew	decision on a premise that is false by commit	only serializable
Phantom	another write changes your search result	SI for read-only queries; write-skew phantoms need index-range locks / serializable

Level	Versions kept	Snapshot per
Read committed	2 (committed + uncommitted)	each query
Snapshot isolation	many committed	whole transaction

Naming chaos	Means
Oracle "serializable"	snapshot isolation
PostgreSQL / MySQL "repeatable read"	snapshot isolation
DB2 "repeatable read"	serializability

Detects lost updates automatically?	
PostgreSQL repeatable read, Oracle serializable, SQL Server snapshot isolation	Yes
MySQL/InnoDB repeatable read	**No**

Serializability	How	Strengths	Weaknesses
Actual serial execution	single thread, stored procedures, in memory	simple; no lock overhead	one core per partition; tiny transactions; cross-partition slow (~1,000 writes/s in VoltDB)
2PL	shared/exclusive locks to end + predicate/index-range locks	decades of use; prevents everything	blocking, unstable tail latency, frequent deadlocks
SSI	SI + conflict detection, abort at commit	no blocking; predictable latency; scales across machines	aborts under contention; long read-write transactions abort

Retrying an aborted transaction — dangers	Why
Commit succeeded, ack lost	runs twice without app-level dedup
Overload	retry worsens it → limit retries, exponential backoff
Permanent error (constraint)	pointless; retry only transient errors
Side effects outside DB (email)	happen even on abort
Client dies mid-retry	data lost

Code to remember

```
UPDATE counters SET value = value + 1 WHERE key = 'foo';   -- atomic write
SELECT * FROM doctors WHERE on_call = true AND shift_id = 1234
  FOR UPDATE;                                               -- lock rows you depend on
UPDATE wiki_pages SET content = 'new'
  WHERE id = 1234 AND content = 'old';                      -- compare-and-set
```

In the interview

Q: What does ACID mean; which letter is misleading? **A:** Atomicity = abortability, consistency = app invariants (app's job), isolation = no interference (often not serializable), durability = committed data survives. The C barely belongs; "ACID compliant" says little.

Q: Read committed vs snapshot isolation? **A:** RC stops dirty reads/writes but allows read skew; SI gives each transaction one consistent snapshot via MVCC, so backups and analytics see a coherent state, and readers never block writers.

Q: Why no read locks for read committed? **A:** One long write would stall many readers.

Q: Lost update — how to prevent? **A:** Atomic updates, `SELECT FOR UPDATE`, automatic detection (PostgreSQL RR, Oracle serializable, SQL Server SI — not InnoDB RR), compare-and-set.

Q: Is compare-and-set always safe? **A:** No — if the WHERE reads from an old snapshot, it may pass despite a concurrent write.

Q: What is write skew? **A:** Two transactions read the same premise and write different objects; together they break an invariant (on-call doctors). SI doesn't catch it; use serializable or lock the rows you read.

Q: Why can't FOR UPDATE fix double-booking? **A:** The check is for rows that do not exist — a phantom; use unique constraints, materialized conflicts, or serializable.

Q: Compare the three serializable implementations. **A:** Serial: simple, one core per partition. 2PL: correct but blocking and deadlock-prone. SSI: optimistic, non-blocking, aborts under contention.

Q: Why does SSI wait until commit to abort? **A:** The reader may be read-only, and the other writer may still abort; waiting avoids needless aborts and keeps long reads working.

Q: Why do apps need to retry aborts, and when is retry dangerous? **A:** Aborts exist so retry is safe (many ORMs don't retry); danger when the commit actually succeeded, under overload, on permanent errors, and with external side effects.

Gotchas

- "Repeatable read" → means different things per vendor; check what you really get.
- Atomicity ≠ concurrency; it is about undoing partial work after a fault.
- Snapshot isolation ≠ serializable → write skew and write-skew phantoms remain.
- MySQL/InnoDB repeatable read → no automatic lost-update detection.
- 2PL latency → unbounded waits; one slow transaction can halt the system.
- Serial execution with interactive transactions → throughput dies on network round-trips; use stored procedures.

TL;DR

Weak isolation leaves races (dirty reads/writes, read skew, lost updates, write skew, phantoms); only serializable stops all of them — via serial execution, 2PL, or SSI.

8. The Trouble with Distributed Systems

Everything that can go wrong once work spans several machines: partial failures, unreliable networks, unreliable clocks, process pauses — and how nodes can know anything at all.

Must-know

- **Partial failure** — some parts broken, others fine, **nondeterministically**; the defining trait of distributed systems. Build **fault tolerance** into software.
- **HPC vs cloud** — supercomputers checkpoint and restart the whole job on a failure (partial → total failure); internet services must stay up on commodity hardware where something is always broken.
- **Asynchronous packet network** — a request can be lost, queued, the remote node crashed or paused, or the request processed and the reply lost or delayed. **Silence looks the same in every case.**
- **Network faults are common** — one medium-sized datacenter study: ~12 faults/month, half a single machine, half a whole rack. **Network partition (netsplit)** = one part cut off from another.
- **Detecting faults** — explicit signals help sometimes (TCP RST/FIN, HBase crash script notifying peers, switch management queries, ICMP Destination Unreachable) but cannot be relied on; you need an **application-level** positive response. TCP ack ≠ processed.
- **Timeouts** — the only practical detector. Too short → false "dead", duplicated actions, **cascading failure** on an overloaded cluster; too long → slow recovery. Choose from measured round-trip distributions or adapt (**Phi Accrual** failure detector: Akka, Cassandra).
- **Unbounded delays** — mostly **queueing**: switch queues, OS queues when CPUs are busy, VM pauses, TCP flow control and retransmission; noisy neighbours in shared clouds.
- **Circuit vs packet switching** — a phone circuit (ISDN: 16 bits per frame, 4,000 frames/s) reserves bandwidth → bounded delay, wasted capacity; Ethernet/IP shares bandwidth for bursty traffic → queues, unbounded delay. Bounded delay is possible but a cost/utilisation trade-off.
- **UDP vs TCP** — VoIP prefers UDP: late data is worthless, so skip retransmission.
- **Time-of-day clock** — wall time since the epoch; NTP-synced; can **jump backwards**; ignores leap seconds → unsuitable for durations.
- **Monotonic clock** — only moves forward (slewed ≤ 0.05%); absolute value meaningless; not comparable across machines → use for timeouts and elapsed time.

- **Clock accuracy is poor** — quartz drift (Google assumes 200 ppm), NTP limited by network delay, VMs pausing, leap seconds, misconfigured NTP, user-controlled device clocks.
- **LWW with timestamps is dangerous** — a causally later write can carry an earlier timestamp and be silently dropped (Cassandra, Riak use LWW). Use **logical clocks** (counters) for ordering.
- **Confidence interval** — a clock reading is really a range; **Spanner TrueTime** returns [earliest, latest] and **waits out the uncertainty** before committing so intervals don't overlap.
- **Process pauses** — GC (sometimes minutes), VM suspend / live migration, context switches and steal time, synchronous disk I/O (even lazy class loading, network disks like EBS), swapping/thrashing, SIGSTOP. A node resumes unaware it was paused.
- **Real-time** — hard real-time needs RTOS, bounded-time libraries, heavy testing; expensive; used in safety-critical embedded systems. Limiting GC impact: planned GC pauses (drain node first) or restart processes periodically.
- **The truth is defined by the majority** — a node cannot trust its own judgement; decisions need a **quorum** (majority: only one can exist at a time).
- **Fencing token** — a number that increases every time a lock/lease is granted (e.g., ZooKeeper zxid/cversion); the **storage service** rejects writes with an older token. Stops a paused ex-holder from corrupting data (the HBase bug).
- **Byzantine fault** — a node lies (sends corrupted or malicious messages). BFT needs **more than two-thirds** of nodes honest; relevant for aerospace and peer-to-peer (Bitcoin), usually not worth it inside one datacenter. Weak lying (corrupted packets, bad input) → checksums, input validation, multiple NTP servers.
- **System models** — timing: synchronous / **partially synchronous** (realistic) / asynchronous; faults: crash-stop / **crash-recovery** (realistic) / Byzantine.
- **Safety vs liveness** — "nothing bad happens" (violation is permanent and pinpointable: uniqueness, monotonic sequence) vs "something good eventually happens" (availability; can carry caveats). **Eventual consistency** is a liveness property.

Key facts & numbers

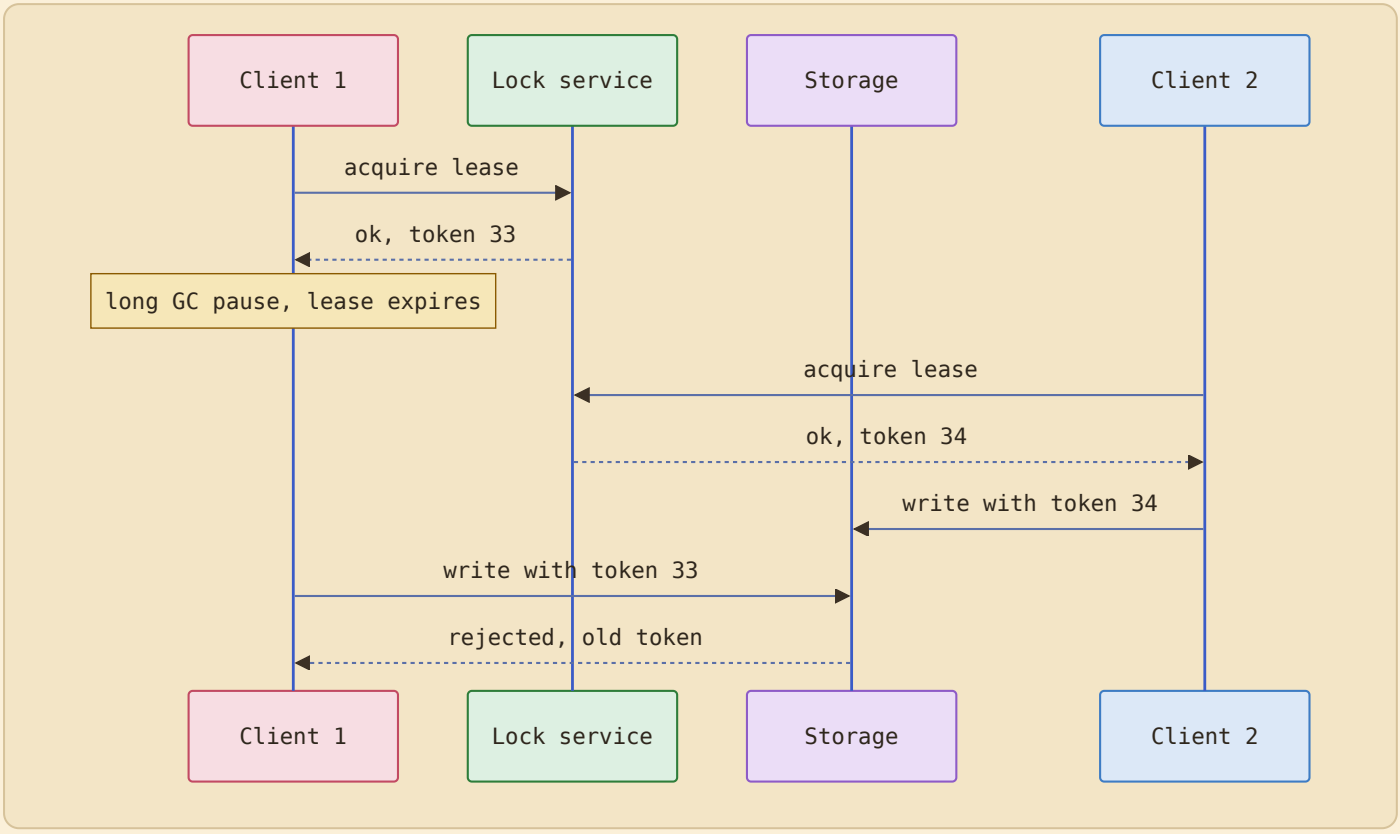
Fact	Value
Network faults in one medium datacenter	~12/month; half single machine, half whole rack
Limping node example	Gigabit NIC dropping to 1 Kb/s (driver bug)
Quartz drift (Google assumption)	200 ppm → 6 ms per 30 s resync, 17 s per day
NTP over internet	~35 ms minimum error, spikes ~1 s
Spanner TrueTime uncertainty	~7 ms (GPS/atomic clock per DC)
Figure 8-3 LWW bug	later write stamped 42.003 vs earlier 42.004, skew < 3 ms
BFT requirement	> 2/3 honest nodes
Majority quorum	3 nodes tolerate 1 failure; 5 tolerate 2

	Time-of-day	Monotonic
Answers	what time is it?	how much time passed?
Linux / Java	`CLOCK_REALTIME` / `System.currentTimeMillis()`	`CLOCK_MONOTONIC` / `System.nanoTime()`
Can jump backwards	yes (NTP reset)	no
Comparable across machines	if synced	no

Timing model	Assumption	Realistic?
Synchronous	bounded delay, pauses, clock error	no
Partially synchronous	usually bounded, sometimes arbitrarily large	**yes**
Asynchronous	no timing assumptions, no clocks	possible but very limiting

Fault model	Assumption
Crash-stop	node crashes and never returns
Crash-recovery	may crash and resume; disk survives, memory lost
Byzantine	may do anything, including lie

	Safety	Liveness
Meaning	nothing bad happens	something good eventually happens
Violation	at a specific moment; cannot be undone	may be satisfied later
Requirement	always, in every situation	may have caveats (majority up, network recovers)



In the interview

Q: Why are timeouts the only practical failure detector, and how do you pick one? **A:** A lost request, dead node, paused node and lost reply all look like silence. Measure round-trip distributions or adapt (Phi Accrual); short → false positives and cascades, long → slow recovery.

Q: Why can't we bound network delay like the phone network? **A:** Packet switching shares bandwidth for bursty traffic, so queues form; reserving circuits wastes capacity and clouds don't offer it.

Q: TCP acked my packet — was it processed? **A:** Not necessarily; the app may crash first. Require an application-level response.

Q: Which clock for a timeout? **A:** Monotonic; time-of-day clocks can jump backwards.

Q: Why is LWW with timestamps dangerous? **A:** Skew can give a causally later write an earlier timestamp, silently dropping it; LWW can't tell sequential from concurrent. Use logical clocks/version vectors.

Q: How does Spanner use clocks safely? **A:** TrueTime gives [earliest, latest]; Spanner waits out the interval before commit, with GPS/atomic clocks keeping it ~7 ms.

Q: How can a lock holder wrongly think it still holds the lease, and how do you fix it? **A:** A process pause (GC, VM suspend, I/O, swapping) lets the lease expire unnoticed. Fix with fencing tokens checked by the storage.

Q: Do fencing tokens stop malicious nodes? **A:** No — that needs Byzantine fault tolerance (> 2/3 honest), usually not worth it in one datacenter.

Q: Which system model do real systems use? **A:** Partially synchronous timing with crash-recovery faults.

Q: Safety vs liveness? **A:** Safety must always hold and a violation is permanent; liveness may be conditional. Eventual consistency is liveness.

Gotchas

- `currentTimeMillis()` for durations → can go backwards; use `nanoTime()`.
- "NTP makes clocks accurate" → only to network-delay precision; misconfiguration can silently drift.
- A single node deciding it is still leader → it may be paused or partitioned; the majority decides.
- Limping node → often worse than a dead one.
- Very short timeouts on overloaded clusters → cascading failure.
- If one machine is enough, stay on one machine — distribution opens Pandora's box.

TL;DR

Networks lose and delay messages, clocks drift and jump, processes pause; so use timeouts carefully, never trust wall clocks for ordering, let the majority decide, and fence stale leaders with tokens.

9. Consistency and Consensus

The strong guarantees you can build on an unreliable network — linearizability, causal ordering, total order broadcast, atomic commit — and why they all come back to consensus.

Must-know

- **Eventual consistency (convergence)** — replicas agree "some time"; no bound, you may not read your own write; bugs appear only under faults or load.
- **Linearizability (atomic / strong consistency)** — the system behaves as if there were **one copy** and each operation takes effect atomically at one instant between request and response. **Recency guarantee**: once any read returns a new value, every later read must return it or newer.
- **Linearizability ≠ serializability** — serializability: multi-object transactions behave as if in *some* serial order; linearizability: single-object real-time recency. Both = **strict serializability** (2PL or actual serial execution); **SSI is not linearizable** (it reads snapshots).
- **Needs linearizability** — **locks and leader election** (ZooKeeper, etcd), **hard uniqueness constraints** (usernames, no negative balance, no double booking), **cross-channel timing dependencies** (photo stored, resize job enqueued, resizer reads old photo).
- **Which replication is linearizable** — single-leader: potentially; consensus: yes; multi-leader: no; leaderless: probably not (LWW with clocks, sloppy quorums; even strict quorums fail under variable delay).

- **CAP** — when the network is **partitioned**, choose **linearizable or available**. "Pick 2 of 3" is misleading (partitions are not optional); the book finds CAP too narrow to be useful.
- **Real cost is latency** — linearizable response time is at least proportional to network-delay uncertainty (Attiya and Welch). Even multi-core RAM gives up linearizability for speed.
- **Causality** — a **partial order** (concurrent ops are incomparable); linearizability is a **total order**. Linearizable ⇒ causally consistent. **Causal consistency** is the strongest model that stays available and fast under network faults.
- **Capturing causality** — track what each operation knew (version vectors); SSI does this for reads.
- **Noncausal sequence generators** (odd/even per node, physical timestamps, block allocators) — break causality.
- **Lamport timestamp** — (counter, node ID); every node and client carries the **max counter seen** → total order consistent with causality. Can't tell concurrency apart, and the order is not **final** at the time you need to decide (unique username).
- **Total order broadcast (atomic broadcast)** — **reliable delivery** + **totally ordered delivery**, order fixed at delivery time = a replicated **log**. Powers **state machine replication**, fencing tokens (zxid), serializable transactions.
- **Linearizable storage from TOB** — append a claim ("I take username X"), wait for it to come back; first claim wins. Linearizable reads: sequence reads through the log (etcd quorum reads), fetch latest position (ZooKeeper `sync()`), or read a synchronously updated replica (chain replication).
- **TOB from linearizable storage** — linearizable increment-and-get counter gives gap-free sequence numbers.
- **Equivalence** — linearizable CAS ⇔ total order broadcast ⇔ consensus (⇔ atomic commit, locks/leases, membership, uniqueness).
- **Atomic commit** — all nodes commit or all abort; single node: the commit record on disk is the deciding moment.
- **Two-phase commit (2PC)** — coordinator gets a global transaction ID; participants do work; **prepare**: each writes data durably and votes yes (promise it *can* commit); coordinator writes decision to its log (**commit point**), then sends commit/abort and **retries forever**.
- **Two points of no return** — participant votes yes; coordinator writes its decision.
- **In doubt (uncertain)** — coordinator crashes after yes votes → participants wait, **holding locks**, until it recovers. If its log is lost → orphaned transactions; manual or **heuristic** resolution (breaks atomicity).
- **3PC** — nonblocking only with bounded delay and a perfect failure detector → not practical.
- **Distributed transactions in practice** — database-internal (VoltDB, MySQL Cluster NDB) often work well; **heterogeneous (XA)** across different systems is hard. MySQL distributed transactions reported >10× slower.
- **XA problems** — coordinator often an unreplicated library in the app server (single point of failure; app no longer stateless); lowest common denominator (no cross-system deadlock detection, no SSI); **failure amplification**.
- **Exactly-once message processing** — atomically commit the message ack and the DB writes via distributed transaction (only if all systems support the same protocol).
- **Consensus properties** — **uniform agreement**, **integrity**, **validity** (safety) + **termination** (liveness; needs a majority alive).
- **FLP** — no deterministic consensus in the asynchronous model; solvable in practice with timeouts or randomness.
- **Algorithms** — VSR, **Paxos** (Multi-Paxos), **Raft**, **Zab**; they are really total order broadcast algorithms.
- **Epochs + quorums** — epoch (ballot / view / term) numbers make the leader unique per epoch; every decision needs a quorum that **overlaps** the election quorum, so a stale leader discovers a higher epoch.

- **Limits of consensus** — synchronous replication cost, strict majority (3 → 1 failure, 5 → 2), fixed membership mostly, timeout sensitivity (frequent elections; Raft edge cases).
- **ZooKeeper / etcd** — small in-memory data, consensus-replicated on 3 or 5 nodes: linearizable atomic ops (locks, leases), **zxid** total order (fencing tokens), sessions + heartbeats + **ephemeral nodes** (failure detection), **watches** (change notifications). Used for leader election, partition assignment, **service discovery**, **membership**. Not for high-rate app state (use e.g. Apache BookKeeper).

Key facts & numbers

Consistency model	Order	Available under partition?	Notes
Eventual	none	yes	only convergence
Causal	partial (cause before effect)	yes	strongest model that stays available
Linearizable	total, real-time	no	single-copy illusion; slow
Serializable	some serial order of transactions	—	not about recency
Strict serializable	serializable + linearizable	no	2PL, actual serial execution

Replication	Linearizable?
Single-leader	potentially (reads from leader or sync followers; if leader is truly the leader)
Consensus (ZooKeeper, etcd)	yes
Multi-leader	no
Leaderless	probably not

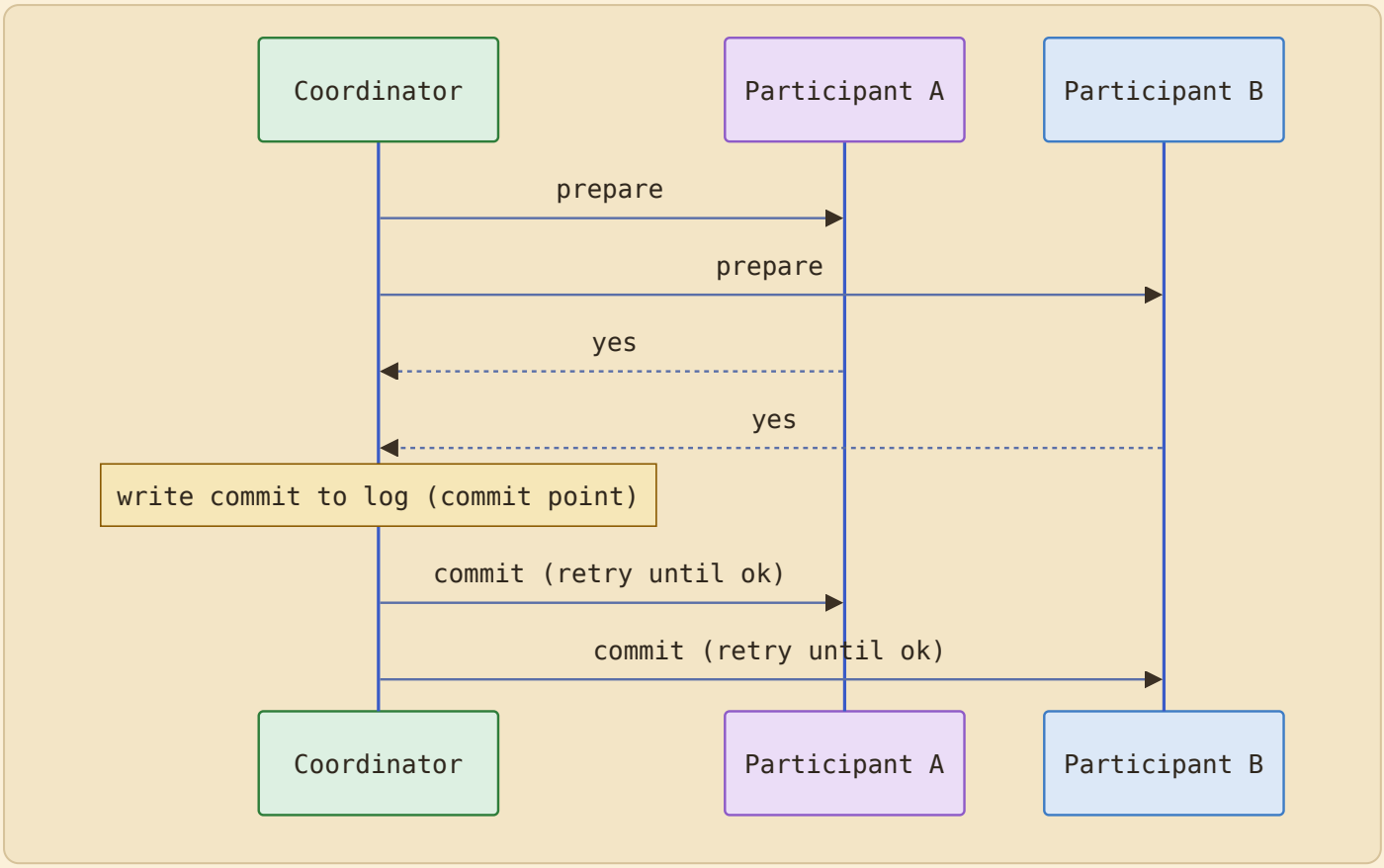
	Version vectors	Lamport timestamps
Purpose	tell concurrent from dependent	impose a total order
Detect concurrency	yes	no
Size	one counter per node	one (counter, node ID) pair

	2PC	Fault-tolerant consensus
Leader	coordinator, not elected	elected per epoch
Votes needed	yes from **every** participant	a **majority**
After failure	wait for coordinator	defined recovery; new leader keeps safety

Consensus property	Type	TOB equivalent
Uniform agreement	safety	same messages, same order everywhere
Integrity	safety	no duplicates
Validity	safety	not corrupted or invented
Termination	liveness	not lost

Epoch name	Algorithm
ballot number	Paxos
view number	Viewstamped Replication
term number	Raft

Leader fails — options	Who does it	Problem
Wait for recovery	many XA/JTA coordinators	violates termination
Manual failover	many relational DBs	human speed
Automatic election	consensus algorithm	use a proven one



In the interview

Q: What is linearizability? **A:** The system acts as if there is one copy and every operation is atomic at an instant between request and response; once anyone reads a new value, all later reads see it.

Q: Linearizability vs serializability? **A:** Recency of single objects in real time vs transactions equivalent to some serial order; both together is strict serializability; SSI isn't linearizable.

Q: Do quorums (w + r > n) give linearizability? **A:** No — with variable delays one reader can see the new value and a later reader the old. Synchronous read repair plus read-before-write fixes reads/writes at a cost, not CAS.

Q: Explain CAP. **A:** During a partition, either refuse requests (linearizable) or serve them (available). Many DBs skip linearizability for performance, not partitions.

Q: How does 2PC work, and what's its weakness? **A:** Prepare → all vote yes → coordinator logs commit → commit retried forever. If the coordinator dies after yes votes, participants are in doubt holding locks.

Q: Why can't in-doubt participants just time out? **A:** Either choice may contradict another participant; only the coordinator knows.

Q: What is consensus? **A:** Nodes propose values and all irrevocably decide one: agreement, integrity, validity, termination.

Q: Isn't consensus impossible (FLP)? **A:** Only in the fully asynchronous model; timeouts or randomness make it solvable.

Q: How do Raft/Paxos avoid split brain? **A:** Epoch numbers + overlapping quorums: a leader can't decide without a quorum that would reveal any newer epoch.

Q: How many nodes to tolerate 2 failures? **A:** Five; four nodes still tolerate only one.

Q: TOB vs linearizability vs consensus? **A:** Equivalent: build CAS from TOB, TOB from a linearizable counter; consensus algorithms are TOB.

Q: Why aren't Lamport timestamps enough for unique usernames? **A:** You never know if another node will later produce a lower timestamp; the order isn't final.

Q: What is ZooKeeper for? **A:** Outsourced consensus: locks/leases, fencing via zxid, failure detection with ephemeral nodes, watches; leader election, partition assignment, service discovery, membership.

Gotchas

- "CAP: pick 2 of 3" → you can't opt out of partitions.
- "Quorum = strong consistency" → not linearizable in general.
- 2PL vs 2PC → isolation vs atomic commit.
- 4-node consensus cluster → no better than 3.
- Heuristic decisions in XA → break atomicity.
- Storing lots of fast-changing data in ZooKeeper → it's for small, slow-changing coordination data.
- Single-leader DB "doesn't need consensus" → it still needs it to choose and hand over leadership.

TL;DR

Linearizability = single-copy recency (costly, not available under partition); causality is the cheaper partial order; total order broadcast, linearizable CAS, atomic commit and consensus are equivalent — outsource them to ZooKeeper/etcd.

10. Batch Processing

Computing derived data from a bounded input: Unix pipelines, MapReduce on a distributed filesystem, join algorithms, and dataflow engines beyond MapReduce.

Must-know

- **System of record vs derived data (Part III intro)** — authoritative copy written first, normalized, wins any disagreement vs data computed from other data (caches, indexes, materialized views, models) that can be rebuilt. A product is not inherently either; it depends on how you use it.
- **Three kinds of systems** — services (online, response time), **batch** (offline, throughput), **stream** (near-real-time, in between).
- **Batch job** — reads **bounded** input, produces derived output, **never modifies its input**; knows when it is done.
- **Unix philosophy** — each program does one thing well; output becomes input of an unknown next program; **uniform interface** (files/stdin/stdout); **separation of logic and wiring**; **transparency** (immutable inputs, inspect any stage).

- **Sort vs in-memory hash** — hash table needs all distinct keys in RAM; `sort` spills to disk and merges (external mergesort, sequential I/O) → scales past memory.
- **HDFS** — shared-nothing distributed filesystem; NameNode tracks blocks; blocks replicated or erasure-coded (Reed–Solomon).
- **MapReduce** — mapper: record → key-value pairs; reducer: all values for one key. One map task per input block, run near the data (**putting computation near the data**); number of reducers configured; reducer for a key chosen by hash of key.
- **Shuffle** — partition mapper output by reducer, sort on mapper's local disk, reducers fetch and merge. Nothing random about it.
- **Workflows** — chained jobs communicate through HDFS directories; schedulers (Oozie, Azkaban, Luigi, Airflow, Pinball); higher-level tools (Pig, Hive, Cascading, Crunch, FlumeJava).
- **Reduce-side sort-merge join** — map both inputs to (join key, record); **secondary sort** lets one side arrive first; reducer joins. Works on any input; costly shuffle.
- **Never query a remote DB per record** — slow, overloads the DB, nondeterministic.
- **Map-side joins** — **broadcast hash join** (small input loaded into each mapper's hash table), **partitioned hash join** (both inputs partitioned identically), **map-side merge join** (also sorted identically). Faster, but need known input properties.
- **Skew / hot keys** — Pig skewed join (sampling job; hot-key records to random reducers, other side replicated), Crunch sharded join (keys declared), Hive skewed join (metadata; map-side join), **two-stage aggregation**.
- **Batch outputs** — build search indexes (Google's original MapReduce use) or **immutable key-value files** bulk-loaded into read-only stores (Voldemort, Terrapin, ElephantDB, HBase bulk load); atomic switch, easy rollback.
- **Human fault tolerance** — immutable inputs + fully replaced outputs → roll back code and rerun.
- **Hadoop vs MPP databases** — storage diversity (dump raw data; **schema-on-read**, **data lake**, "sushi principle: raw data is better"), processing diversity (any code: MapReduce, Hive, Spark, HBase, Impala), and designing for frequent faults (task-level retry, eager disk writes).
- **Dataflow engines** (Spark, Tez, Flink) — whole workflow as one DAG of operators; sort only when needed; intermediate state in memory/local disk; pipelined; reuse JVMs; recover via lineage (Spark **RDD**) or checkpoints (Flink).
- **Deterministic operators** — recomputed data must match what downstream consumed; avoid unordered hash iteration, unseeded randomness, clocks, external lookups.
- **Graph processing** — **Pregel / bulk synchronous parallel**: "think like a vertex"; rounds of messages; vertex state in memory; checkpoint between iterations (Apache Giraph, Spark GraphX, Flink Gelly). Partitioning by arbitrary vertex ID causes heavy cross-machine messaging.
- **High-level APIs** — declarative joins let a **cost-based optimizer** choose algorithms; **vectorized execution**; domain libraries (Mahout, MADlib, k-nearest neighbours).

Key facts & numbers

	Reduce-side join	Map-side join
Input assumptions	none	known size / partitioning / sort order
Cost	sort + copy + merge; may hit disk several times	no reducers, no sort
Output layout	partitioned and sorted by join key	follows the large input

Join	When
Sort-merge (reduce-side)	default; any inputs
Broadcast hash	one input small enough for memory (e.g. 10 TB clicks × 200 MB products)
Partitioned hash	both inputs partitioned the same way
Map-side merge	partitioned **and** sorted the same way

	MapReduce workflow	Dataflow engine
Scheduling unit	independent jobs chained by directories	one job, DAG of operators
Intermediate state	HDFS, replicated	memory or local disk
Sorting	always between map and reduce	only where needed
Next stage starts	after all previous tasks finish	as soon as input is ready
Recovery	re-run task from durable input	recompute from lineage / checkpoint
Process startup	new JVM per task	reused JVMs

	MPP database	MapReduce
Node crash	abort whole query	retry just that task
Memory vs disk	in memory where possible	eager to write to disk
Best for	short queries, SQL	huge long jobs likely to hit a task failure

Fact	Value
Google preemption	an hour-long task has ~5% chance of being killed (> 10× hardware failure rate)
100 tasks × 10 min	> 50% chance at least one is killed
Distinct URLs, small/mid site	fit in ~1 GB RAM
Graph fits on one machine	single-machine algorithm (or GraphChi on disk) likely beats distributed

Code to remember

```
cat /srv/logs/web-access.log |
  awk '{print $7}' |      # URL field
  sort | uniq -c |        # count identical URLs
  sort -r -n | head -n 5  # top 5
```

In the interview

Q: Explain MapReduce. **A:** Mappers turn records into key-value pairs near the data; the shuffle partitions by key hash, sorts and delivers each partition to a reducer, which processes all values per key; output goes to HDFS; inputs never change, so tasks retry safely.

Q: Why is there always a sort? **A:** It makes equal keys adjacent so reducers use little memory, and external merge sort scales past RAM with sequential I/O.

Q: How do you join two large datasets? **A:** Reduce-side sort-merge join by default; broadcast hash join if one side is small; partitioned hash or map-side merge join if inputs are laid out identically.

Q: How do you handle a hot key? **A:** Detect it (sampling), spread its records across random reducers and replicate the other side, or do two-stage aggregation.

Q: How should batch results reach a serving DB? **A:** Build immutable DB files in the job and bulk-load them into a read-only store with atomic switch — not per-record writes (slow, overloads DB, breaks all-or-nothing output).

Q: MapReduce vs Spark/Flink? **A:** MapReduce materializes every intermediate result to replicated HDFS and always sorts; dataflow engines run a DAG, pipeline in memory, sort only when needed, and recover by recomputation.

Q: Why does determinism matter? **A:** A recomputed partition that differs forces downstream operators to be rerun too.

Q: Why was MapReduce so eager to write to disk? **A:** At Google, low-priority batch tasks are often preempted, so task-level retry from durable data avoids rerunning whole jobs.

Q: Hadoop vs MPP databases? **A:** MPP: monolithic, schema up front, fast SQL, abort whole query on failure. Hadoop: any data format and processing model, task retry. They are converging.

Gotchas

- Calling a remote DB per record from a mapper → slow and nondeterministic; copy the DB into HDFS and join.
- Writing to a production DB from reducers → partial, duplicated side effects.
- Nondeterministic operators in dataflow engines → recovery cascades.
- Distributing a graph that fits in RAM → often slower than one machine.
- "DAG" in dataflow engines ≠ graph data.

TL;DR

Batch = immutable bounded input → derived output; MapReduce partitions, sorts and retries tasks; joins are sort-merge, broadcast or partitioned; dataflow engines keep the model but drop needless materialization.

11. Stream Processing

Batch processing done continuously on unbounded input: message brokers and logs, change data capture, event sourcing, time and windows, stream joins, and exactly-once.

Must-know

- **Event** — small, self-contained, immutable record of something that happened, with a timestamp. **Producer** writes it once; many **consumers** read it; related events grouped into a **topic** (stream).
- **Polling vs notification** — polling gets expensive at low latency; messaging systems notify consumers.
- **Producer faster than consumer** — **drop**, **buffer**, or **backpressure** (Unix pipes, TCP). Also ask: what if nodes crash — are messages lost?
- **Direct messaging** (UDP multicast, ZeroMQ, StatsD, webhooks) — low latency, but the app must handle loss.
- **Message broker** — a database optimised for streams: deletes messages once delivered, small working set, topic subscription instead of queries, notifies on change.
- **Load balancing vs fan-out** — each message to one consumer (shared work) vs to all consumers; combinable with consumer groups.
- **Acks and redelivery** — broker redelivers unacked messages; with load balancing this **reorders** messages (fix: a queue per consumer).
- **Log-based broker** (Kafka, Kinesis, DistributedLog) — append-only **partitioned log** on disk; each partition read in order by one consumer in a group; progress = **offset** (like a replication-log

position); messages retained → **replay**. Parallelism ≤ number of partitions; a slow message blocks its partition (**head-of-line blocking**).

- **Disk as buffer** — a large but bounded **circular buffer**; a slow consumer falls behind and only it is affected; monitor lag.
- **Dual writes** — writing app data to DB and index separately **races** (DB ends at B, index at A) and can partially fail → permanent silent inconsistency.
- **Change data capture (CDC)** — make the DB the leader and turn its replication log into a stream consumed by indexes, caches, warehouses (followers). Tools: Debezium/Maxwell (MySQL binlog), Bottled Water (PostgreSQL WAL), Mongoriver (oplog), LinkedIn Databus, Facebook Wormhole. Asynchronous → replication-lag anomalies apply.
- **Initial snapshot** — tied to a known log offset; or **log compaction** (keep latest value per key; null = tombstone) so a new consumer can start from offset 0 of a compacted topic.
- **Event sourcing** — store immutable **application-level events** (intent) as the source of truth; derive current state; snapshots for speed. Not compactable per key.
- **Commands vs events** — a command may fail validation; once accepted it becomes an immutable **event** (fact) that consumers cannot reject → validate synchronously before writing, or split into tentative + confirmation events.
- **State = integral of the stream; stream = derivative of state** — the log is the truth, the DB is a cache of the latest log values.
- **Immutability advantages** — auditability (accounting ledger), recovery from bugs, more information kept (item added then removed from cart), **several read views from one log** (**CQRS**: separate write and read forms).
- **Limits of immutability** — high churn → big history; **deletion** for legal/privacy reasons needs rewriting history (Datomic excision, Fossil shunning); copies linger in backups and SSDs.
- **Uses** — **CEP** (queries stored, events flow through them: Esper, IBM InfoSphere Streams), **stream analytics** (rates, rolling averages, trends; probabilistic algorithms like Bloom filters, HyperLogLog), **materialized views**, **search on streams** (percolator), message passing.
- **Event time vs processing time** — when it happened vs when the processor saw it. Processing-time windows create artifacts (restart backlog looks like a spike).
- **Stragglers** — you never know a window is complete: ignore late events (monitor drops) or publish a **correction**.
- **Untrusted device clocks** — log time occurred (device clock), time sent (device clock), time received (server clock); offset = received − sent; apply to occurred.
- **Windows** — tumbling, hopping, sliding, session (see table).
- **Stream joins** — stream-stream (window join), stream-table (**enrichment** from a CDC-maintained local copy), table-table (**materialized view** of the join). Order across streams isn't guaranteed → nondeterministic; version joined data (slowly changing dimensions), at the cost of log compaction.
- **Exactly-once / effectively-once** — records may be processed twice but the visible effect is once. **Microbatching** (Spark Streaming, ~1 s), **checkpointing** with barriers (Flink), **atomic commit** internal to the framework (Google Cloud Dataflow, VoltDB, Kafka planned), **idempotent writes** (store Kafka offset with the value).
- **Rebuilding state** — remote datastore (slow per message), local state replicated (Flink snapshots to HDFS; Samza/Kafka Streams changelog topic; VoltDB redundant processing), or replay input.

Key facts & numbers

	AMQP/JMS broker	Log-based broker
Delivery	individual messages to consumers	whole partition to one consumer, fixed order
Progress	ack each message; deleted after ack	checkpoint offset
Parallelism	per message	per partition
Reread	no	yes (retained on disk)
Best for	task queue: expensive messages, order irrelevant	high throughput, fast messages, order matters, derived data
Examples	RabbitMQ, ActiveMQ, HornetQ, Qpid, TIBCO, IBM MQ, Azure Service Bus, Google Cloud Pub/Sub	Kafka, Amazon Kinesis, Twitter DistributedLog

	Batch (Ch 10)	Stream (Ch 11)
Input	bounded, known size	unbounded, never complete
Storage abstraction	distributed filesystem	broker / event log
Output	written at the end, all-or-nothing	emitted continuously
Fault recovery	discard failed task output, rerun	microbatch, checkpoint, transactions, idempotence
Time	event timestamps in data, no clock reads	event vs processing time, stragglers

	CDC	Event sourcing
DB use	mutable; app unaware	built on immutable events
Events mean	low-level state changes (full new record)	app-level intent
Later event overrides earlier?	yes	usually no
Log compaction	yes	no — full history needed

Window	Definition	Example
Tumbling	fixed length, each event in exactly one	10:03:00–10:03:59, then 10:04:00–10:04:59
Hopping	fixed length, overlapping	5 min with 1-min hop: 10:03:00–10:07:59, 10:04:00–10:08:59
Sliding	events within an interval of each other	5 min: 10:03:39 and 10:08:12 together
Session	per user, closes after inactivity	ends after 30 min idle

Join	Inputs	State kept	Example
Stream-stream	two event streams (or self-join)	recent events of both sides	searches + clicks by session ID within 1 h
Stream-table	events + DB changelog	local copy of the table	enrich activity with user profiles
Table-table	two changelogs	latest state of both	Twitter timelines from tweets + follows

Fact	Value
Disk buffer	6 TB at 150 MB/s fills in ~11 hours; in practice days or weeks

In the interview

Q: Kafka-style log or RabbitMQ-style queue? **A:** Queue for task-style work where messages are expensive and order doesn't matter; log for high-throughput ordered streams, replay and derived data.

Q: Why does a queue reorder messages? **A:** Load balancing plus redelivery after a consumer crash sends the message to a consumer that has moved on.

Q: What limits parallelism in a log? **A:** Consumers in a group ≤ partitions; one slow message blocks its partition.

Q: How do you keep a search index/cache in sync with the DB? **A:** Not dual writes — CDC: the DB is the leader, its log is published in order, index and cache consume as followers; bootstrap from a snapshot at a known offset or a compacted topic.

Q: Event sourcing vs CDC? **A:** Immutable app-level events as the source of truth vs low-level changes extracted from a mutable DB; event logs can't be compacted per key and need snapshots.

Q: How do you validate a seat booking with events? **A:** Validate synchronously before emitting the event, or split into tentative reservation + confirmation events.

Q: Event time vs processing time, and late data? **A:** Window by event time; ignore stragglers (and monitor) or publish corrections; correct device clocks with sent/received offsets.

Q: Name the window types. **A:** Tumbling, hopping, sliding, session.

Q: Name the stream joins. **A:** Stream-stream (window), stream-table (enrichment), table-table (materialized view); all keep state from one side and query it with the other.

Q: How do stream processors get exactly-once? **A:** Microbatching or checkpoints inside the framework; for external effects, internal atomic commit or idempotent writes (needs deterministic replay order, no concurrent writers, fencing on failover).

Q: Is exactly-once literally true? **A:** No — effectively-once: repeated processing, one visible effect.

Gotchas

- Dual writes → race and partial failure; use CDC or event log.
- Windowing by processing time → restart backlog looks like a traffic spike.
- Too many consumers for a Kafka topic → extras idle (max = partitions).
- CDC consumers read your own writes? → no guarantee; it's async.
- Immutable log + "right to be forgotten" → needs excision-style rewriting.
- Stream joins with changing dimension data → nondeterministic unless versioned.

TL;DR

Streams are unbounded batch: log-based brokers give ordered, replayable partitions; CDC and event sourcing turn databases into streams; handle time with event-time windows and stragglers; get effectively-once with checkpoints, atomic commit or idempotence.

12. The Future of Data Systems

Putting it all together: integrating specialised tools through derived data, "unbundling" the database, correctness end to end, and doing the right thing with data.

Must-know

- **Data integration** — no single tool fits every use, so apps compose several; make one store the **system of record** and **derive** everything else (indexes, views, caches, ML models) from its ordered change log.
- **Dual writes are the enemy** — two stores may apply conflicting writes in different orders; a single log decides the order.
- **Derived data vs distributed transactions** — log position instead of locks; deterministic retry + idempotence instead of atomic commit; asynchronous (no read-your-writes by default) instead of linearizable; faults stay local instead of amplifying.
- **Limits of total ordering** — throughput beyond one leader (partitioned logs), multiple datacenters, microservices with their own state, offline clients. Causality across systems (unfriend then message) needs logical timestamps or logging "what the user saw".
- **Reprocessing for evolution** — rerun new code over the whole history to build a new view; run old and new views side by side and shift traffic gradually (**gradual, reversible migration**) — like a railway gauge change with a third rail.
- **Lambda architecture** — batch layer (Hadoop) + speed layer (Storm) over the same immutable events; problems: logic maintained twice, merging outputs, costly reprocessing. **Unified** engines need replay, exactly-once, and **event-time** windows (Apache Beam on Flink / Cloud Dataflow).
- **Unbundling databases** — a database's internals (indexes, materialized views, replication logs, triggers) done as separate systems connected by logs. `CREATE INDEX` = snapshot + catch up on backlog, just like adding a follower.
- **Federation vs unbundling** — **unifying reads** (one query interface over many stores, polystore; PostgreSQL foreign data wrappers) vs **unifying writes** (CDC + event logs feed every store). Still missing: a high-level "shell" to compose them.
- **Unbundled vs integrated** — unbundling gives breadth; if one product meets all needs, use it (fewer moving parts, more predictable performance).
- **Application code as a derivation function** — secondary index, full-text index, ML model, cache are all functions of the source data.
- **Stream processors and services** — instead of a synchronous call to an exchange-rate service, subscribe to the rate stream and keep a local copy: faster and more robust.
- **Write path vs read path** — eager work when data arrives vs lazy work at query time; the derived dataset is where they meet. Caches/indexes/materialized views **move the boundary**.
- **Push to clients** — offline-capable clients hold a cache of server state; push state changes to devices (EventSource, WebSockets) → end-to-end event streams (Elm, React/Flux/Redux). **Reads are events too** (log reads to capture causality; costs storage).
- **Exactly-once end to end** — TCP dedup, DB transactions, stream-processor exactly-once each cover only one hop; a user resubmitting a timed-out POST looks new. Fix with a client-generated **operation ID** + uniqueness constraint.
- **End-to-end argument** (Saltzer, Reed, Clark, 1984) — a function can be fully correct only with the endpoints' knowledge; low-level features help performance but don't suffice (checksums,

encryption, dedup).

- **Uniqueness needs consensus** — but it scales by partitioning: log requests partitioned by hash of the value; a single-threaded processor per partition decides in order and emits success/rejection.
- **Multi-partition without atomic commit** — log the whole request as one message (with request ID), a deterministic processor splits it into debit/credit messages, per-account processors dedupe by ID.
- **Timeliness vs integrity** — see table; integrity usually matters more. Dataflow systems give up guaranteed timeliness but keep integrity with exactly-once/idempotence.
- **Loosely interpreted constraints** — many business constraints can be violated temporarily and fixed with a **compensating transaction** (apology, refund, overdraft fee); the apology process must exist anyway.
- **Coordination-avoiding systems** — integrity without atomic commit or linearizability; geo-distributed, multi-leader; serializable transactions only where needed.
- **Trust, but verify** — hardware corrupts data, software has bugs (MySQL uniqueness, PostgreSQL write skew under serializable bugs); HDFS and S3 continually re-read and compare replicas. **Auditability**: event-sourced, deterministic dataflow allows hashing logs and rerunning derivations; Merkle trees, certificate transparency.
- **Doing the right thing** — predictive analytics can lead to bias, discrimination and "**algorithmic prison**" (exclusion without proof or appeal); accountability; **feedback loops** (bad credit score → can't get a job → worse score). Tracking becomes **surveillance**; consent is weak when terms are opaque and opting out isn't a real choice. **Privacy** = the right to decide what to reveal to whom — surveillance transfers that right to the collector. Data is a hazardous asset (breaches, future governments). Industrial Revolution analogy: regulation eventually fixed pollution and child labour.

Key facts & numbers

	Distributed transactions	Log-based derived data
Write order	locks (2PL)	position in a log
Exactly-once via	atomic commit	deterministic retry + idempotence
Timing	usually linearizable (read your writes)	asynchronous
Faults	one participant aborts all; failure amplification	stay local; slow consumer falls behind
Across vendors	XA: poor fault tolerance and performance	simple, heterogeneous

	Timeliness	Integrity
Meaning	users see the latest state	no loss, duplication or contradiction; derived data matches source
Violation	temporary ("eventual consistency")	permanent ("perpetual inconsistency")
Tools	linearizability, read-after-write	atomicity, durability, exactly-once, idempotence
Example	card statement a day behind: fine	money disappearing: not fine

	Federation (unify reads)	Unbundling (unify writes)
Idea	one query interface over many stores	every change reliably reaches every store
Example	PostgreSQL foreign data wrapper	CDC + event logs
Tradition	relational: one integrated system	Unix: small tools + pipes

Search option	Write-path work	Read-path work
No index (grep)	none	huge
Normal index	update per term	look up terms + Boolean logic
Precompute all queries	infinite	trivial
Precompute popular queries (materialized view)	update cached results	fast for common queries

Loose constraint	Compensation
Same username/seat grabbed twice	apologise, choose another
Oversold stock	reorder, apologise, discount
Airline/hotel overbooking	refunds, upgrades, neighbouring hotel
Overdrawn account	overdraft fee; daily withdrawal limit bounds risk

Code to remember

```
ALTER TABLE transfer_log ADD UNIQUE (op_id);
BEGIN TRANSACTION;
INSERT INTO transfer_log (op_id, payer, payee, amount)
  VALUES ('9b1f...f501', 88, 77, 25.00);  -- fails on retry
UPDATE wallets SET funds = funds + 25.00 WHERE wallet_id = 77;
UPDATE wallets SET funds = funds - 25.00 WHERE wallet_id = 88;
COMMIT;
```

In the interview

Q: Keep a DB, search index and cache consistent? **A:** One system of record; capture its changes in order (CDC/event log); each derived store consumes the log with idempotent, deterministic consumers; replay rebuilds any store.

Q: Why not XA/2PC across them? **A:** Poor fault tolerance and performance, failure amplification, no standard across heterogeneous systems; the log contains faults.

Q: What do you give up? **A:** Timeliness — no read-your-writes by default; clients can wait for the change if needed.

Q: Lambda vs unified engine? **A:** Lambda duplicates logic in batch and speed layers and must merge outputs; unified engines replay history with exactly-once and event-time windows.

Q: Make a payment exactly-once end to end? **A:** Client-generated operation ID carried through every hop, unique constraint where the effect is recorded; 2PC doesn't cover a user resubmitting a timed-out POST.

Q: Enforce unique usernames in a log-based system? **A:** Partition requests by hash of the name; a single-threaded processor per partition checks and emits accept/reject; scales with partitions.

Q: Timeliness vs integrity? **A:** Temporary staleness vs permanent corruption; integrity matters more, and dataflow keeps it without coordination.

Q: When NOT to unbundle? **A:** When one product already covers your needs.

Q: Why audit an ACID database? **A:** Hardware corrupts data and software has bugs; ACID assumes bug-free transactions. Verify end to end.

Q: What does privacy mean per the book? **A:** The right to decide what to reveal to whom; surveillance moves that right to the collector.

Gotchas

- Dual writes to DB + index → permanent divergence.
- "Exactly-once" in one component → not end-to-end; use operation IDs.
- Lambda architecture → two codebases for one logic.
- App-level "check then insert" for uniqueness → breaks under weak isolation; use a unique constraint.
- Strict linearizable constraints everywhere → often unnecessary when compensation exists anyway.
- Blind trust in the database → audit and verify.

TL;DR

Compose specialised stores by deriving everything from ordered logs; unbundle the database; guarantee integrity end to end with operation IDs, idempotence and async checks; and remember the people behind the data.

13. Glossary — terms people mix up

The book's glossary, cut to the pairs that interviewers use to test precision.

Must-know

- **Atomic** — two meanings: (1) concurrency: appears to happen at one instant; (2) transactions: all writes committed or all rolled back.
- **Consistency** — four meanings: replica convergence (eventual consistency), consistent hashing, linearizability (CAP), ACID invariants.
- **Skew** — two meanings: uneven load (hot spots) or a timing anomaly (read skew, write skew, clock skew).
- **Linearizable** — behaves as if one copy changed by atomic operations. **Serializable** — concurrent transactions give the same result as some serial order.
- **2PC** — all nodes commit or all abort. **2PL** — locks held to end of transaction for serializable isolation.
- **Idempotent** — safe to retry: running it several times = once.
- **Backpressure = flow control** — slow the producer when the receiver can't keep up.
- **Bounded / unbounded** — known maximum size or delay vs none.
- **Total vs partial order** — every pair comparable vs some pairs incomparable (concurrent).
- **System of record vs derived data** — authoritative source written first vs repeatable transformation that can be rebuilt.
- **Materialize** — compute eagerly and store the result.
- **Split brain** — two nodes both think they are leader.
- **Quorum** — minimum number of nodes that must vote for an operation to succeed.
- **Timeout** — simplest fault detector; cannot tell node failure from network failure.
- **Shared-nothing** — independent nodes with own CPU, memory, disks, linked by an ordinary network.
- **Stored procedure** — a transaction's whole logic runs on the DB server, no client round-trips.

TL;DR

When a word has two meanings (atomic, consistency, skew), say which one you mean.

Last-minute checklist

- Fault ≠ failure; hardware faults are independent, software bugs systematic, humans the top cause of outages.
- Describe load with parameters (fan-out!) and performance with percentiles measured client-side; never average percentiles.
- Documents for trees, relational for joins, graphs for many-to-many; "schemaless" = schema-on-read.
- LSM = memtable + SSTables + compaction + Bloom filters (writes); B-tree = pages + WAL + splits (reads); 4 levels × 500 × 4 KB = 256 TB.
- OLTP = seek-bound key lookups; OLAP = bandwidth-bound scans → star schema, column store, compression, sort order.
- Backward = new reads old; forward = old reads new; protobuf tags are forever; Avro needs defaults and the writer's schema.
- Single-leader / multi-leader / leaderless; async failover can lose writes; fence split brain.
- Read-your-writes, monotonic reads, consistent prefix — know one fix for each.
- w + r > n overlaps sets but is not linearizable; sloppy quorum + hinted handoff trade consistency for availability.
- Concurrent = neither knew of the other; version vectors detect it; LWW silently loses data.
- Partition by range (scans, hot spots) or hash (even, no ranges); rebalance by moving whole partitions, never mod N.
- Local index = scatter/gather reads; global index = async multi-partition writes; routing via ZooKeeper or gossip.
- Isolation ladder: read committed (no dirty reads/writes) → snapshot isolation (no read skew; MVCC) → serializable (no write skew/phantoms).
- Lost update fixes: atomic ops, FOR UPDATE, auto-detection (not InnoDB RR), compare-and-set.
- Serializable = serial execution (one core), 2PL (blocking; not 2PC!), SSI (optimistic; aborts at commit).
- Networks lose/delay messages; timeouts are the only detector; TCP ack ≠ processed.
- Monotonic clock for durations; wall clocks jump; Spanner waits out TrueTime uncertainty (~7 ms).
- The majority decides; fencing tokens stop paused ex-leaders; BFT needs > 2/3 honest.
- Linearizable = one copy + recency; causal = partial order, strongest that stays available; CAP only during partitions.
- Linearizable CAS ⇔ total order broadcast ⇔ consensus; Raft/Paxos/Zab use epochs + overlapping quorums; 5 nodes tolerate 2 failures.
- 2PC: two points of no return (yes vote, coordinator decision); coordinator crash → in doubt, locks held.
- Batch: bounded immutable input; sort-merge, broadcast hash, partitioned hash joins; dataflow engines skip needless materialization.
- Stream: log brokers (offsets, replay, partitions) vs queues (acks); CDC over dual writes; event time + windows; effectively-once.
- Future: system of record + derived views via logs; end-to-end operation IDs; integrity over timeliness; verify, and treat data with humanity.